Can Model Confidence Triage Qualitative Coding?

Decision Models, LLMs, and What a 5% Error Guarantee Actually Costs

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Abstract

Language models can apply qualitative codebooks, but their labels save effort only if a researcher knows which ones to trust. We ask whether model confidence can tell, comparing seven decision models (APIs that return probabilities for typed questions) with twelve LLM confidence signals on five human-coded tasks spanning Reddit comments, counselling dialogue and classroom talk, under a prospectively specified analysis plan. Three findings stand out. First, low calibration error does not make confidence useful: decision models differ widely in how well they rank their own mistakes, and calibration can be improved without improving ranking. Second, error targets are hard to keep. A threshold tuned to 5% error on hand-coded comments exceeds 5% on new comments in about half of re-splits, and applying an independence-based guarantee to clustered decisions breaks it, because a comment's codes and a transcript's utterances err together. A guarantee that respects how the hand-coded sample was drawn holds, at a price: after 30% of Reddit comments are hand-coded, it certifies 60–83% of the remaining code decisions for most coders. On dialogue, with whole transcripts hand-coded, it certifies nothing and could not even for a perfect coder at our calibration sizes; sampling single utterances instead still yields almost no certificates. Third, accepted shares overstate the work saved: nearly all certified decisions say a code does not apply, and a baseline that reads no text accepts 75% of comment decisions at 3.6% error. Model uncertainty also rises where annotators disagree, and option order changes some models' labels. Confidence can triage coding, but only if researchers check ranking, sampling design and what the accepted labels are. Models were called on 10–12 October 2026; we release the harness and every coding decision.

1 Introduction

Deductive qualitative coding applies a fixed codebook to text: each segment of an interview, dialogue or survey answer is read, compared with the code definitions and labelled, often twice to establish reliability [29]. Large language models (LLMs) can now apply codebooks with useful accuracy [7, 11, 17, 39, 65, 73], but accuracy alone does not tell a researcher which of a model's labels to trust. The practical promise is *triage*: accept the labels the model is confident about, and send the rest to a person.

Consider a researcher with 1,500 counselling utterances to code and time to check 400 by hand. A model labels every utterance with a confidence, and a threshold decides which labels to accept. Whether this works depends on three properties of the confidence that are easy to conflate. *Error ranking*: do the model's mistakes receive lower confidence than its correct labels? *Calibration*: does a stated 90% mean 90% correct? *Selective risk and coverage*: at the chosen threshold, how many labels are accepted, and how many of those are wrong? The three can come apart. A well-calibrated model can rank its errors poorly, and a threshold chosen on the 400 hand-coded utterances to give 5% error can exceed 5% on the other 1,100. We ask whether model confidence can select low-error decisions reliably enough for this workflow.

A new class of products is marketed for exactly this purpose. We call them *decision models*: interfaces that take typed questions (yes/no, multiple choice) with criteria and return a probability for each option,

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without generating text. TypeSafe's Jev [57] was followed within weeks by OpenAI's Decisions API [42], Cloudflare's Clef [8], Perplexity's pplx-decider [45] and the open Kev models [43]. The interface is new, but the idea behind it, probabilistic classification with the option to abstain, is not [16]. Whether these probabilities support triage on real codebooks, with human disagreement, dialogue context and rare codes, is an open question. We study it for the classification step of deductive coding on short segments (comments, utterances, sentences); inductive coding and long-form interviews are outside our scope.

We compare seven decision models with twelve LLM confidence signals (verbalized confidence, token probabilities and agreement-based signals) on five human-coded tasks: moral foundations in Reddit comments, client and therapist talk in counselling dialogues, and teacher and student moves in K-12 mathematics lessons. A prospectively specified analysis plan fixes three research questions, each answered with pre-specified contrasts:

- **RQ1 (triage).** Do decision models' probabilities rank their own coding errors better than LLM confidence signals, and what share of decisions can be accepted at $\leq$5% error?
- **RQ2 (disagreement).** Is model uncertainty high where human coders disagree, and how does a model's agreement with annotators compare with annotators' agreement with each other?
- **RQ3 (robustness).** How much do decisions change when the codebook or request is reworded or reordered, or when context is removed?

The research questions say what happens; six experimental questions ask why, and what it means in practice. Three examine the answer to RQ1: whether low calibration error predicts useful triage (**EQ1**), whether threshold rules keep their error target and what a finite-sample guarantee costs (**EQ2**), and whether triage is limited by poor ranking or by large ties at the top of the confidence scale (**EQ3**). **EQ4** tests one explanation for decision models' advantage, by comparing two 9B decision models with the confidence of a 9B instruct model from the same family. **EQ5** turns the results into a practical choice: which configurations give the most guaranteed automation for their cost, and how much reading they actually save. **EQ6** checks whether the results are artefacts of nondeterminism, prompt choice, memorization or label noise.

The short answer to our title is: on Reddit comments, yes, with caveats; on dialogue, not at the budgets we studied. Two facts drive both halves. Coding decisions come in clusters, such as the eight codes of one comment or the utterances of one transcript, so an error guarantee has to follow how the hand-coded sample was drawn. And most decisions in a multi-label codebook are easy calls that a code does not apply, so a high accepted share can mean little.

Relation to concurrent work. Two studies released in the weeks before ours also evaluate decision models: Ibrahim and Zaki [25] compare Jev and small open decision models with 19 LLMs on 15 computational-social-science labelling tasks (accuracy, calibration, coverage at fixed thresholds, cost and cascades), and TypedBench [51] uses synthetic, rule-labelled tasks. Where the designs overlap, our findings agree: Jev trails the strongest LLMs in accuracy, its raw calibration beats most LLM signals but not the frontier models, and cascades save cost. Risk-controlled triage is not new either: HyPAC [70] routes items between models and humans with a PAC guarantee on annotation error, and Zhao and Liu [72] triage qualitative codes by confidence and inter-model diversity. Our contribution is an evaluation, not a new method, and it has four parts.

Contributions.

1. **A multi-vendor evaluation** (all questions; §3). Seven decision models and twelve LLM confidence signals on five human-coded tasks, under a prospectively specified plan with dated amendments and a deviations log, with every coding decision released.
2. **Ranking, not calibration, separates coders** (RQ1, EQ1, EQ3, EQ4; §4.1, §5.1–§5.4). Decision models range from ranking their errors worse than a cheap LLM's verbalized confidence to nearly as well as a frontier LLM's. Calibration error does not predict where a model falls: task-level recalibration improves calibration while leaving the ranking of decision models and verbalized confidence unchanged, and per-code recalibration narrows most of the differences. A comparison of related 9B systems suggests that much of a decision model's advantage comes from reading out a probability rather than from its training.
3. **Guarantees that respect the sampling design** (RQ1, EQ2; §3.4, §5.2). Tuning a threshold to 5% error on hand-coded comments overshoots in about half of re-splits, and a guarantee that treats decisions

as independent fails because decisions within a comment or transcript err together. A corpus-level rule, adapted from betting tests for sampling without replacement, holds: after 30% of comments are hand-coded, it certifies 60–83% of the remaining decisions for most coders. On dialogue, with whole transcripts hand-coded, it certifies nothing. We show why: with whole transcripts as sampling units, our procedure could not certify even a perfect coder at our calibration sizes, and sampling single utterances removes that obstacle but leaves too little evidence of $\leq$5% error among the coders' confident decisions.

4. **What automation means for coding** (EQ5; §5.5, §6). Accepted shares overstate the work saved. Nearly all certified comment decisions say a code does not apply, a baseline that reads no text accepts 75% of comment decisions at 3.6% error, and under every certified policy a person still reads most comments. Recall of applying codes, the share of segments still read and policy-specific costs give a fuller picture, and we turn them into a recipe for researchers.

Beyond triage, model uncertainty is higher where human coders disagree (RQ2, §4.2); option order and missing context change some models' labels substantially (RQ3, §4.3); and the checks of EQ6 bound the main threats: an alternative prompt does not improve the LLM baselines, there is little evidence of memorization, and the contrasts barely change against an adjudicated reference, although adjudication also suggests errors in some human labels (§5.6). Section 2 reviews related work, Section 3 describes the design and the threshold rules, Sections 4 and 5 answer the research and experimental questions, and Section 6 discusses what they mean for researchers.

2 Related work

LLMs for qualitative and content coding. LLMs have been used to apply codebooks deductively [7, 65], to replace or assist crowd annotation [17, 73], to code open-ended survey responses [40] and to score classroom discourse [60], and Dunivin [12] showed that chain-of-thought prompting can match human coders on some hermeneutic codes. Most of this work asks how well a model agrees with human coders, and best-practice guidance accordingly stresses validation against human labels [44, 55]. Recent studies refine the workflow: they compare prompting strategies [50], benchmark many models against expert adjudication [38], design human–AI workflows for deductive coding [34, 39, 68] and add model self-critique [13]. Fewer ask which of a model's labels can be trusted. Zhao and Liu [72] come closest: they combine self-confidence with inter-model diversity to route segments to review, auto-accepting 35% at under 5% error on their data. We ask whether such thresholds hold on held-out data and across codebooks, and what it takes to guarantee them. Agreement with human labels is itself an imperfect yardstick: Liu et al. [36] show that blind experts often prefer the model's code to the human consensus, and our adjudication points the same way (§5.6).

Decision models. Jev introduced the "System One" interface [57]. Denney and DiGiuseppe [10] compare Jev with LLMs on seven political-science replications for accuracy, cost and calibration. Li et al. [33] use it as an accept-or-escalate judge, and Rafe and Das [48] convert crash narratives into probabilistic variables. Ibrahim and Zaki [25] and TypedBench [51] are closest to our work (§1). Other recent studies show that short, natural-sounding context can flip decision models' answers [67], that rejection thresholds fitted on one dataset fail on another [37], and that decisions can be read from an ordinary LLM's next-token probabilities without retraining, with option-order bias reduced by averaging over rotations [32, 71]. Relative to this work, we add the newer hosted and open models [8, 42, 43, 45], real qualitative codebooks with human disagreement, a test of whether thresholds keep their error targets, guarantees that respect how the hand-coded sample was drawn, and a comparison of related 9B systems.

Confidence and calibration. Neural classifiers are often miscalibrated, and temperature scaling [20], Platt scaling [47] and isotonic regression [69] are standard fixes. Whether confidence separates a model's correct from incorrect predictions is a separate question [22], and it is the one triage depends on. For LLMs, confidence can be read from token probabilities [27] or elicited verbally [54, 66], and fast black-box estimators are being developed [21]. We compare these signals with decision models' probabilities and keep calibration and error ranking apart throughout (§5.1).

Selective prediction and error guarantees. Selective classification accepts a prediction only above a confidence threshold [16], and cascades send the rest to a stronger model [6]. Two kinds of guarantee exist for such thresholds. Conformal risk control [1] bounds an expected monotone loss under exchangeability; standard Learn-then-Test [3] gives a high-probability bound on population risk from valid tests, under assumptions such as i.i.d. calibration data. When observations come in groups, the group becomes the unit of analysis [14], and for a fixed corpus a finite-population betting test [62] needs only that the hand-coded units, and the order they are visited in, are random. These guarantees have been applied to LLM judges [5, 26, 31], hybrid LLM–human annotation [70], human verification of LLM labels [61] and calibrated grading [49]. We adapt the finite-population test to coding decisions, averaging over visiting orders with e-values [59], and measure three things on real codebooks: how often the usual rule without a guarantee breaks its target, whether a guarantee that ignores clustering fails, and how much coverage a valid one costs at calibration sizes a research team can realistically hand-code.

Human label variation and downstream inference. Annotators disagree for principled reasons, so a single majority label is an imperfect target [46, 58]. Baan et al. [4] argue that calibration should then be judged against the distribution of human labels, and Lee et al. [30] test whether LLMs reproduce that distribution. Our RQ2 asks the triage version of this question: whether model uncertainty is high where humans split. A complementary line of work does not accept individual labels at all. It corrects downstream estimates, such as code prevalences or regression coefficients, using a hand-coded subsample [2, 15], and can use model confidence to choose which items to hand-code [18].

Contamination. Public benchmarks may be in training data. We adapt the guided-versus-general completion test of Golchin and Surdeanu [19] and add a decoy-name control to separate memory from task framing.

3 Study design

3.1 Tasks and codebooks

We use five tasks from three public, human-coded datasets (Table 1).

- **MFRC** [56]: Reddit comments coded for moral foundations with the dataset's guidelines. The codebook is multi-label with 8 codes; 1,988 of the 2,000 sampled comments have three annotators and 12 have two.
- **AnnoMI** [63, 64], full version: transcribed motivational-interviewing demonstrations, with mutually exclusive codes for client talk (change, sustain, neutral) and therapist behaviour (reflection, question, input, other). AnnoMI publishes labels rather than full definitions, so its codebooks combine AnnoMI's labels with MITI 4 wording [41] and the AnnoMI-to-MISC mapping of Sun et al. [52]. A subset is coded by ten annotators.
- **TalkMoves** [53]: K-12 mathematics classroom transcripts coded with the dataset's manual, with mutually exclusive teacher moves (six plus none) and student moves (four plus none).

Together the tasks vary what makes coding hard: a sparse multi-label codebook against exclusive ones, self-contained comments against utterances that depend on the previous turn, and low against moderate agreement among human coders (Krippendorff's α of 0.17 on MFRC, 0.47–0.74 on AnnoMI). For dialogue tasks the coder also sees the previous turn. Dunivin's coded passages about W. E. B. Du Bois [12] serve as a replication task. We draw stratified samples of 2,000 comments (MFRC) or 1,500 segments (each dialogue task) that keep rare codes, and weight each segment by its inverse inclusion probability [24]. A hash of the cluster id (comment, transcript or lesson) assigns about 30% of clusters to a calibration split and the rest to a test split; because clusters differ in size, the calibration split holds 25–40% of segments. The reference label is the annotator majority; ties are dropped.

3.2 Coders and confidence

Table 2 lists the models. **Decision models** receive identical requests: the segment and its context, and either one multiple-choice question (exclusive codebooks) or one yes/no question per code (multi-label), whose

Table 1: Tasks, codebooks and samples. Calib and test are segments; clusters are conversations or transcripts (comments for MFRC). Dunivin is the replication task.

task	unit	codebook	codes	calib	test	clusters	annot./item	human α
MFRC	Reddit comment	multi-label	8	608	1392	2000	3	0.17
AnnoMI client	client utterance	exclusive	3	569	931	128	1 (10 on a subset)	0.47
AnnoMI therapist	therapist utterance	exclusive	4	602	898	132	1 (10 on a subset)	0.74
TalkMoves teacher	teacher sentence	exclusive	7	370	1130	444	1	–
TalkMoves student	student sentence	exclusive	5	383	1117	436	1	–
Dunivin	historical passage	multi-label	9	34	86	120	1	–

Multi-label: one yes/no decision per code; exclusive: one choice among the codes (counts include any 'none' or 'other' option). Dialogue units include the previous turn as context. α: Krippendorff's alpha on multiply-coded items (macro over codes for multi-label); – where items are single-coded. RQ2 uses MFRC and AnnoMI's ten-annotator subset.

criteria are the code's definition, inclusion and exclusion rules and examples. OpenAI Decisions receives the same content mapped to its own choice and predicate types; Clef, pplx-decider and Kev accepted Jev's requests unchanged. We self-host Kev on Modal GPUs from the authors' release. **LLMs** receive the same codebook text in one prompt and return JSON with a one-sentence rationale, an answer and a probability from 0 to 100 (per code for multi-label tasks). Appendix D shows both requests.

From outputs to confidence. For decision models, an exclusive decision is the most probable option and its confidence is that probability; a multi-label decision says a code applies if its probability is at least 0.5, with confidence $\max(p, 1-p)$. LLM signals are derived as follows.

- *Verbalized, exclusive:* the chosen option gets the stated probability (at least $1/K$), and the other options share the remainder equally. RQ2's model entropy for these signals therefore comes from a reconstructed, not an elicited, distribution.
- *Verbalized, multi-label:* we read the stated probability as confidence in the stated yes/no answer. Read as "probability that the code applies", it contradicted the answer for 85% of Luna's code decisions (0.2–4.6% for the other LLMs). Confidence is therefore $c = \max(q, 1-q)$, and $p(\text{applies})$ is c for "yes" and $1-c$ for "no".
- *Token probability:* the probability of the answer token, read after the generated rationale: the option letter for exclusive codebooks and the yes/no token for each multi-label code (Luna, gpt-oss and both Qwen models).
- *3-LLM agreement:* the vote share of the Luna, Haiku and gpt-oss verbal runs; *self-consistency:* the vote share of five Luna samples at temperature 1 on 350 segments per task.

We run reasoning models with minimal reasoning so that answer-token probabilities exist. Sonnet and Qwen3.8-2.4T are the frontier references; Qwen3.8-2.4T runs on a 270-segment-per-task subset. Every model was called on 10–12 October 2026 (UTC), several of them in public beta (Table 36 lists the versions).

3.3 Decisions and metrics

A *decision* is one label to accept or reject: one per segment for exclusive codebooks, one per segment and code for multi-label ones. **Error AUROC** (RQ1, primary) is the probability that a correct decision has higher confidence than an incorrect one, with ties counted as one half [22]: 0.5 is chance, and 1 means every error has lower confidence than every correct decision. **Coverage@5%** (RQ1, primary) is the share of test decisions accepted by the threshold rule below, reported with the error among them. Secondary metrics are the area under the risk–coverage curve (AURC), expected calibration error (ECE, 10 bins), Cohen's κ [9] and cost. RQ1 and EQ1–EQ5 use design weights, as do RQ3's changes; RQ2's correlations and leave-one-annotator-out κ are unweighted. Pooled estimates are equal-weight means over the five tasks; this describes these five tasks, not a population of codebooks. Confidence intervals come from a cluster bootstrap that resamples calibration and test clusters separately and refits thresholds in every replicate (1,000 replicates), unless stated otherwise.

Table 2: Models and confidence signals, with realized API cost per 1,000 segments (October 2026 prices).

model	kind	confidence signals	sample	USD / 1k segments
Jev	decision model	probability	full	0.047
OpenAI Decisions	decision model	probability	full	0.102
Clef	decision model	probability	full	0.213
Clef-flash	decision model	probability	full	0.080
pplx-decider	decision model	probability	full	0.019
Kev-9B	open decision model	probability	full	0.073
Kev-27B	open decision model	probability	full	0.315
Luna	LLM	verbalized; token prob.; self-consistency (350/task, $0.28)	full	0.107
Haiku	LLM	verbalized	full	1.906
gpt-oss	LLM	verbalized; token prob.	full	0.232
Qwen3.5-9B	LLM	verbalized; token prob.	full	0.176
Sonnet	LLM (frontier)	verbalized	full	1.964
Qwen3.8-2.4T	LLM (frontier)	verbalized; token prob.	270/task	2.481
3-LLM agreement	ensemble	vote share	full	2.245

Kev: estimated serving cost from observed throughput on Modal (Kev-9B 30 calls/s on 2 H100s at $3.95/h; Kev-27B 11 calls/s on 2 B200s at $6.25/h), excluding cold starts and idle time. Model identifiers, settings, versions returned and call dates are in Table 36 (appendix).

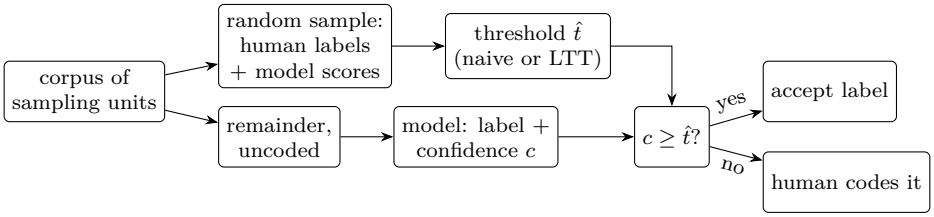


Figure 1: The triage workflow we evaluate. A random sample of the corpus's units (here comments, transcripts or lessons) is coded by hand and by the model; comparing the two fixes a threshold. The model's confidence then decides which labels in the rest of the corpus are accepted and which go to a person. A guaranteed rule bounds the error rate among the accepted labels of the remainder.

3.4 Threshold rules

A researcher who has hand-coded a sample needs a rule that turns it into a threshold. Each rule below picks a confidence threshold t from the hand-coded calibration clusters and accepts the other decisions with confidence $\geq t$; they range from common practice to a rule with a guarantee. We distinguish two targets. The *corpus target* is the design-weighted error among the accepted decisions of the uncoded remainder of the corpus, the quantity a researcher coding their own data cares about. The *population target* is the expected error among accepted decisions for new clusters from the same source. Whether one particular test set has error above 5% is exactly the corpus target when that test set is the uncoded remainder of a random split, and only a diagnostic for the population target.

- **Naive** (the pre-specified primary rule): the lowest threshold whose calibration error is at most 5%. No guarantee.
- **LTT on decisions:** Learn-then-Test [3] with binomial tests of "risk >5%" on decision counts, fixed-sequence from the most confident tie block. With design weights the counts are Kish-effective and fractional, so the test is an approximation even for independent decisions. It treats decisions as independent; we show it only for comparison.
- **Corpus LTT** (our guaranteed rule). Intuitively, the test gathers evidence, cluster by cluster, that the accepted decisions are wrong less than 5% of the time, and certifies a threshold only when that evidence is strong enough to extend to the rest of the corpus. Formally, for each threshold on a grid of 16 values from 0.5 to 1.0, cluster i contributes $Z_i(t) = \sum_j w_{ij}\,\mathbf{1}\{c_{ij} \geq t\}\,(e_{ij} - 0.05)$, so a set of clusters has error at most 5% among its accepted decisions exactly when its Z-sum is at most 0. Rescaled by B, the largest total weight of one cluster in the corpus (known before anything is coded), $Y_i = (Z_i + 0.05B)/B$ lies

in $[0, 1]$. A betting test for sampling without replacement [62] visits the hand-coded clusters in random order and asks whether the corpus mean of Y is low enough for the remainder's Z-sum to be at most 0 (Appendix B). So that the result does not depend on one arbitrary order, we average the test's final wealth over 100 random orders. Each threshold is tested at 0.05/16, and the lowest certified one is used.

Proposition (corpus LTT; Appendix B). Suppose the hand-coded clusters are a uniformly random subset of the corpus's clusters, the orders in which the test visits them are uniformly random and independent of the labels, and the confidence scores, the threshold grid and B are fixed before the hand-coded labels are used. Then, with probability at least 0.95 over the random choice of clusters and orders, every certified threshold has error at most 5% among the accepted decisions of the uncoded clusters (counting a remainder with no accepted decision as within the target). No assumption is made about how errors depend on each other within or across clusters.

The construction adapts existing testing machinery to the coding workflow; the statistical tools are not new. Nothing in it requires the unit to be a cluster: any unit can be sampled, such as a comment, a whole transcript, a fixed-size block of turns or a single utterance coded with its context, provided the hand-coded units are a random sample of the corpus's units and B is the largest unit's weight. What the guarantee must respect is how the hand-coded sample was drawn.

How we apply it. In our evaluation, each task's sample is the corpus, its calibration split the hand-coded clusters (assigned by a hash of the cluster id, which we treat as random) and its test split the remainder. The visiting orders come from a pseudo-random generator with a seed recorded before this analysis was run. The guarantee holds for each configuration on its own; picking the best of several certified configurations would need a further correction. Learned scores, such as per-code maps or the prevalence baseline, are fitted on half of the calibration clusters, frozen, certified on the other half and applied unchanged to the remainder. Multi-stage cascades are certified jointly over all combinations of per-stage thresholds.

Two reference rules. For the population target we also report the same test with a constant null mean (*LTT on i.i.d. clusters*). It needs clusters to be independent draws, which we assume rather than know, and B to bound every future cluster's weight, which is true for MFRC (eight codes per comment at one weight) but not for transcripts of unbounded length. The **oracle** applies the naive rule to the test split itself: it is not deployable, and serves as an upper bound for comparison.

3.5 Analysis plan and reference labels

Analysis plan. The plan fixed the tasks, splits, decision units, metrics, primary rule and RQ1 contrasts before any test-split outcome was computed. Four amendments, each dated before the data it governs, added OpenAI Decisions and a Qwen comparator (v1.1), five more decision models (v1.2), a memorization probe (v1.3) and a prompt-sensitivity check (v1.4). The plan and amendments were kept in the project repository, not registered with an external service, so their timing cannot be verified independently; we call the plan prospectively specified. A claim needs the pooled 95% CI to exclude zero; for the v1.2 family the plan also requires Holm's correction [23], which we apply to the earlier families as a sensitivity analysis. Jev anchors the confirmatory contrasts because the original plan was written for it; the other decision models were added by amendment. Pooled intervals resample each corpus's clusters jointly, so AnnoMI's two tasks share transcript draws and TalkMoves' two share lesson draws. Sonnet was specified on a nested subset; that result is the confirmatory one, and its later full-sample run is a labelled extension. Everything changed after results were seen is labelled exploratory and logged (Appendix A); this includes the corpus guarantee, which replaced decision-level LTT and was revised after external reviews.

Adjudicated reference. Human majority labels can contain errors. We sampled 200 items for blinded adjudication from disagreements between the human label and six label sources (Jev in both question formats, Luna, Haiku, gpt-oss and Sonnet's subset run), oversampling high-confidence disagreements, plus hidden controls. A two-family LLM panel (Claude Opus 5.5 and GPT-6 Sol) judged each item. The author adjudicated 36 of the 52 items the panel disputed, plus a 12-item audit of panel-agreed items (agreement 10 of 12); a planned second sitting to measure the author's own consistency was not done. Of the 200 items, 184

were resolved; unresolved items keep the human label. We use this reference as a sensitivity analysis, not as a corrected gold standard.

4 Results: research questions

Table 3 gives a short answer to each of the nine questions. This section gives the evidence for the three research questions, and Section 5 for the six experimental ones. Each subsection starts with the answer, then gives the evidence and the reasons behind it, and ends with what it means for a researcher.

Table 3: The questions and their short answers.

question	short answer	§
RQ1: Does confidence triage coding?	Partly. Ranking quality varies widely across decision models (error AUROC 0.69–0.79; Sonnet's verbalized confidence 0.80), and how much can be accepted depends mostly on the task.	4.1
RQ2: Is uncertainty high where humans disagree?	Yes, moderately, for most coders; no coder falls below the range of individual annotators.	4.2
RQ3: How stable are decisions?	Mostly stable to rewording, less so to option order; removing context changes 15–24% of dialogue labels on average.	4.3
EQ1: Does calibration error predict triage?	No. Low calibration error does not imply good ranking, and calibration can be improved without improving ranking.	5.1
EQ2: Do threshold rules keep their target?	The usual rule overshoots in about half of re-splits. A corpus guarantee holds, certifying 60–83% of comment decisions for most coders; on dialogue, none with whole transcripts hand-coded and almost none with utterance sampling.	5.2
EQ3: Ranking or ties at the top?	Both. Large, error-prone ties limit Jev and OpenAI Decisions; per-code maps split ties and improve ranking for every coder with enough data.	5.3
EQ4: Do related 9B decision models beat a 9B instruct model?	Against its verbalized confidence, yes; against its token probabilities, barely.	5.4
EQ5: What gives the most guaranteed automation per dollar?	pplx-decider per inference dollar; several decision-model configurations come within two points of Sonnet at a fraction of its cost, but a person still reads most comments.	5.5
EQ6: Are the results artefacts?	The checks bound the main threats; adjudication suggests errors in some human labels.	5.6

4.1 RQ1: Does confidence triage qualitative coding?

Answer: partly. Confidence ranks coding errors well above chance for most coders, but how well depends on the coder more than on the interface, and how much can be accepted depends on the task more than on either. Table 4 summarizes agreement and ranking for every signal, Table 5 gives the confirmatory contrasts, and Figures 2 and 3 show the contrasts and the per-task risk–coverage curves.

Decision models differ widely in how well they rank their errors. The open Kev-27B ranks its own errors best among the decision models (error AUROC 0.788), close to Sonnet's verbalized confidence (0.801). pplx-decider (0.764) and Jev (0.756) follow; OpenAI Decisions, Clef, Clef-flash and Kev-9B rank errors worse (0.694–0.736). In the pre-specified contrasts, Jev ranks its errors better than four of the six other decision models: OpenAI Decisions by +0.020 [0.005, 0.035], Clef by +0.036, Clef-flash by +0.045 and Kev-9B by +0.062. Its difference from pplx-decider is not significant (−0.008 [−0.027, 0.010]), though pplx-decider accepts more decisions under the naive 5%-target rule (−0.079 [−0.245, −0.047]). Kev-27B beats Jev on both primary metrics (−0.032 AUROC; −0.113 coverage).

Jev beats most LLM confidence signals, but not the strongest verbalized ones. Jev ranks its errors better than every token-probability signal, by 0.20 AUROC for Luna and gpt-oss and by 0.07 for the Qwen models, and better than 3-LLM agreement (+0.087), self-consistency (+0.153) and gpt-oss's verbalized confidence (+0.029). The verbalized confidence of Haiku (−0.026) and Sonnet beats it: −0.048 [−0.082, −0.015] on the pre-specified Sonnet subset ($p = 0.002$) and −0.045 [−0.060, −0.032] on the full-sample extension. Jev's advantage over Luna's verbalized confidence is borderline (+0.016 [0.0004, 0.031], bootstrap $p = 0.046$) and not a claim after Holm adjustment, and the exploratory Qwen3.8-2.4T's verbalized confidence edges out Jev (−0.033 [−0.069, −0.001]). In our rationale-first read-out, token probabilities concentrate near 1 and rank errors poorly: 96–100% of their decisions carry confidence ≥0.99, and for Luna more than half are tied at exactly 1.0, which no threshold can split. EQ4 examines whether the read-out contributes. What drives the differences among the decision models themselves is the subject of EQ1, EQ3 and EQ4: calibration, ties at the top of the scale and the read-out.

How much can be accepted depends on the task, and on what is being accepted. Under the naive rule, Kev-27B, Sonnet and pplx-decider accept 41–44% of all decisions and Jev 33% (Table 6; equal-weight means over the five tasks), but 6.2–7.5% of what they accept is wrong. Acceptance is far higher on MFRC, where these four accept 78–94% of decisions, than on the four dialogue tasks (22–32% on average). The difference is mostly about what the decisions are. On MFRC, 89% of test decisions are codes that do not apply, and most of those are easy: a prevalence baseline that reads no text, predicting "absent" for every code with confidence equal to that code's calibration share of absent labels, accepts 75% of MFRC decisions at 3.6% error under the naive rule (Table 7), although it carries no certificate (§5.5). The models differ in what the baseline cannot do, which is to find the codes that apply: among accepted decisions, pplx-decider labels 62% of the applying codes correctly, Sonnet 53% and Jev 35%, and the rest go to a person or are missed. Dialogue decisions, by contrast, are choices between neighbouring codes, such as a reflection and a question, and there the naive rule's error targets mostly fail (§5.2).

Robustness. Against the adjudicated reference, 39 of the 40 Jev contrasts (20 comparators, two metrics) keep their sign and significance. The exception is the borderline error-ranking contrast with Luna's verbalized confidence, whose interval moves from just above zero to just below it (+0.016 [−0.0001, 0.032]).

What it means. "Decision model" names an interface, not a level of quality: ranking has to be measured on the researcher's own hand-coded sample, and a strong LLM's verbalized confidence is a competitive alternative. A high accepted share is not evidence of a useful coder by itself; it should be compared with a baseline that reads nothing and with the share of applying codes found. And the naive rule's 5% is a target, not a guarantee: the leading coders' accepted decisions are wrong 6–7.5% of the time, which Section 5.2 explains.

Table 4: RQ1: agreement, calibration and error ranking on the test split (mean over the five tasks, design-weighted). Groups: decision models, verbalized LLM confidence, token probabilities, agreement signals.

signal	sample	κ (weighted)	ECE	error AUROC	AURC	conf $\geq$ 0.99
Jev	full	0.475	0.120	0.756	0.144	0.220
OpenAI Decisions	full	0.449	0.122	0.736	0.155	0.309
Clef (27B)	full	0.377	0.055	0.721	0.228	0.000
Clef-flash (9B)	full	0.369	0.065	0.712	0.234	0.000
pplx-decider	full	0.482	0.052	0.764	0.122	0.084
Kev-9B	full	0.300	0.100	0.694	0.281	0.000
Kev-27B	full	0.474	0.073	**0.788**	0.105	0.003
Luna verbal	full	0.483	0.180	0.741	0.146	0.226
Haiku verbal	full	0.465	0.154	0.783	0.149	0.003
gpt-oss verbal	full	0.452	0.204	0.727	0.192	0.043
Qwen3.5-9B verbal	full	0.383	0.303	0.646	0.276	0.389
Sonnet verbal	full	0.529	0.051	**0.801**	0.108	0.026
Qwen3.8-2.4T verbal	270/task	0.499	0.102	0.802	0.122	0.002
Luna logprob	full	0.483	0.243	0.558	0.225	0.990
gpt-oss logprob	full	0.452	0.284	0.558	0.264	0.998
Qwen3.5-9B logprob	full	0.384	0.334	**0.684**	0.237	0.964
Qwen3.8-2.4T logprob	270/task	0.499	0.248	0.703	0.148	0.961
3-LLM agreement	full	0.493	0.153	**0.670**	0.193	0.719
Luna self-consistency	350/task	0.506	0.196	0.624	0.197	0.851

Error AUROC: probability that a correct decision is more confident than an incorrect one (ties count half). AURC: area under the risk-coverage curve, lower is better. conf $\geq$ 0.99: share of decisions with confidence $\geq$ 0.99. Bold: the highest error AUROC among full-sample signals in each group; Table 5 tests the differences.

Table 5: Confirmatory contrasts, Jev minus each comparator X, pooled over the five tasks (paired cluster bootstrap, thresholds refitted in every replicate). Positive values favour Jev. Coverage is the naive rule's share accepted.

family	X	Δ err. AUROC [95% CI]	Holm p	better	Δ cov@5% [95% CI]	Holm p	better
v1.0	Luna verbal	+0.016 [0.0004, 0.031]	0.184	–	+0.128 [-0.044, 0.180]	0.510	–
v1.0	Haiku verbal	-0.026 [-0.043, -0.011]	0.016	X	+0.168 [-0.071, 0.184]	0.510	–
v1.0	gpt-oss verbal	+0.029 [0.011, 0.051]	0.016	Jev	+0.108 [-0.064, 0.155]	0.510	–
v1.0	Luna logprob	+0.198 [0.177, 0.221]	0.016	Jev	+0.329 [0.148, 0.381]	0.016	Jev
v1.0	gpt-oss logprob	+0.198 [0.174, 0.221]	0.016	Jev	+0.317 [0.148, 0.367]	0.016	Jev
v1.0	3-LLM agreement	+0.087 [0.069, 0.103]	0.016	Jev	+0.329 [0.160, 0.382]	0.016	Jev
v1.0	Luna self-consistency	+0.153 [0.118, 0.186]	0.016	Jev	+0.164 [0.160, 0.446]	0.016	Jev
v1.0	Sonnet verbal (v1.0 subset)	-0.048 [-0.082, -0.015]	0.016	X	-0.249 [-0.296, -0.029]	0.110	–
v1.1	OpenAI Decisions	+0.020 [0.005, 0.035]	0.024	Jev	+0.103 [-0.050, 0.187]	0.274	–
v1.2	Clef (27B)	+0.036 [0.017, 0.054]	0.010	Jev	+0.090 [-0.085, 0.135]	1.000	–
v1.2	Clef-flash (9B)	+0.045 [0.024, 0.064]	0.010	Jev	+0.075 [-0.086, 0.128]	1.000	–
v1.2	pplx-decider	-0.008 [-0.027, 0.010]	1.000	–	-0.079 [-0.245, -0.047]	0.010	X
v1.2	Kev-9B	+0.062 [0.041, 0.086]	0.010	Jev	+0.090 [-0.041, 0.158]	0.872	–
v1.2	Kev-27B	-0.032 [-0.047, -0.017]	0.010	X	-0.113 [-0.272, -0.063]	0.010	X
v1.1 (sec.)	Qwen3.5-9B verbal	+0.110 [0.093, 0.129]	–	Jev	+0.295 [0.149, 0.364]	–	Jev
v1.1 (sec.)	Qwen3.5-9B logprob	+0.072 [0.050, 0.093]	–	Jev	+0.236 [0.065, 0.283]	–	Jev
ext.	Sonnet verbal (full sample)	-0.045 [-0.060, -0.032]	–	X	-0.105 [-0.283, -0.055]	–	X

'better' names the winner (Jev, or X for the comparator) when the pooled 95% CI excludes 0 and Holm p < 0.05 within the family; – : no claim. Holm is part of the plan's rule for v1.2 and a sensitivity analysis for v1.0 and v1.1. Sonnet enters v1.0 as specified, on its nested subset; the full-sample extension (ext.) is exploratory and outside the family. v1.1 secondary rows are unadjusted. Intervals resample each corpus's clusters jointly for related tasks; coverage intervals are asymmetric because refitted thresholds jump between tie blocks.

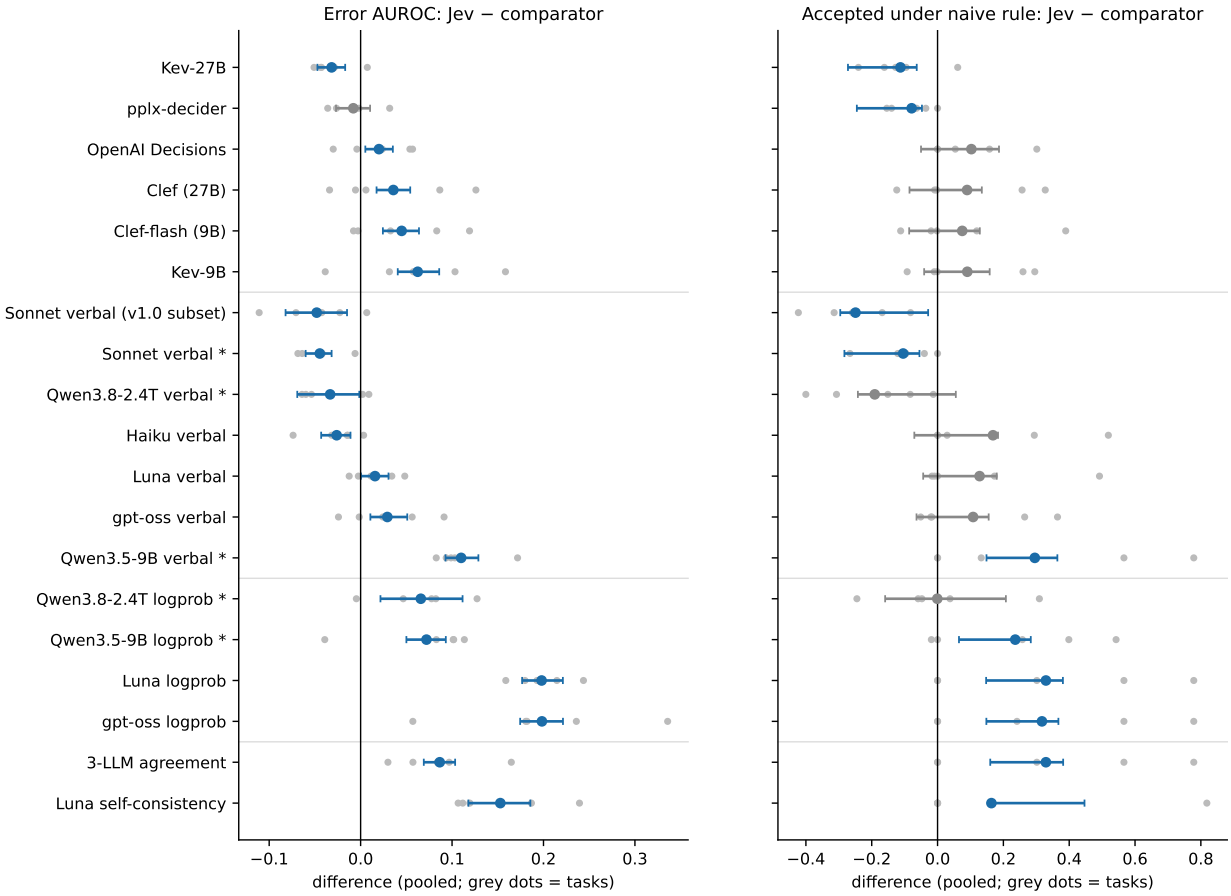


Figure 2: RQ1 contrasts: Jev minus each signal, pooled over tasks (grey dots: individual tasks), grouped as decision models, verbalized confidence, token probabilities and agreement signals. Positive values favour Jev. Blue marks a 95% CI that excludes 0 (unadjusted). Left: error AUROC. Right: share accepted under the naive 5%-target rule. *: secondary or exploratory comparison (the Qwen models; Sonnet on the full sample, an extension of its pre-specified subset).

Table 6: Threshold policies at a 5% error target on the frozen split (percent): share of test decisions accepted, and error among them. Pooled columns average the five tasks; dialogue columns average the four dialogue tasks.

signal	naive: accepted	naive: error	oracle: accepted	MFRC: LTT on decisions	MFRC: corpus LTT	its error	dialogue: naive	dialogue: LTT on decisions	dialogue: corpus LTT
Jev	32.9	**6.8**	16.3	74.4	59.8	1.6	21.7	0.0	0.0
OpenAI Decisions	22.7	4.7	17.8	65.6	0.0	–	10.2	0.0	0.0
Clef (27B)	24.0	**5.8**	20.3	88.4	82.3	3.0	7.4	0.0	0.0
Clef-flash (9B)	25.4	**7.0**	20.7	86.7	77.9	2.7	9.5	0.0	0.0
pplx-decider	40.8	**7.5**	32.1	88.5	77.1	2.3	28.0	0.0	0.0
Kev-9B	23.9	**6.3**	19.2	82.8	67.1	2.4	8.1	0.0	0.0
Kev-27B	44.2	**7.0**	37.2	91.3	76.6	2.1	31.7	10.6	0.0
Luna verbal	20.2	4.6	16.2	79.5	64.0	2.2	5.3	0.0	0.0
Haiku verbal	16.1	3.2	16.8	74.9	65.8	1.7	1.4	0.0	0.0
gpt-oss verbal	22.1	**5.6**	20.6	0.0	0.0	–	6.9	0.0	0.0
Sonnet verbal	43.4	**6.2**	40.1	87.5	82.7	2.5	31.7	6.1	0.0
Qwen3.8-2.4T verbal	37.9	4.3	38.0	74.8	0.0	–	26.8	0.0	0.0

Naive: threshold with calibration error $\leq 5\%$ (the pre-specified rule). Oracle: fitted on test, an unattainable upper bound. LTT on decisions treats decisions as independent and is shown only for comparison; corpus LTT certifies the error of the uncoded remainder (Section 3.4). Bold: naive-rule error above the 5% target.

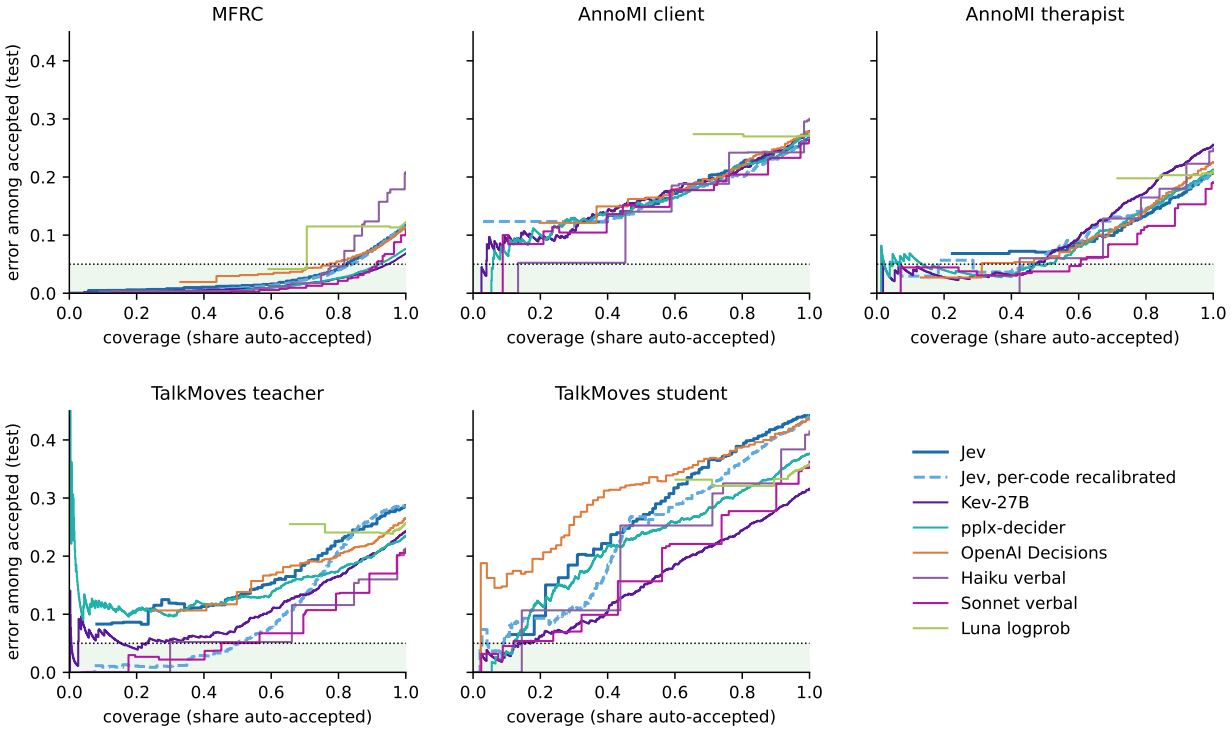


Figure 3: Risk–coverage on the test split for a subset of coders. Each curve accepts decisions from most to least confident. The dotted line is 5% error, and the shaded band is the region a 5% target accepts. On the dialogue tasks most coders never reach it at any meaningful coverage.

Table 7: MFRC: what the accepted decisions are (percent). 11% of test decisions are codes that apply. The prevalence baseline reads no text: it predicts 'absent' for every code, with confidence equal to that code's share of 'absent' labels.

coder	naive: accepted	naive: error	naive: applying codes found	corpus LTT: accepted	corpus LTT: error	corpus LTT: share 'absent'	corpus LTT: applying codes found	corpus LTT: applying codes deferred
Jev	77.8	3.8	34.6	59.8	1.6	96.6	17.1	75.8
OpenAI Decisions	72.5	4.4	12.9	0.0	–	–	0.0	100.0
Clef (27B)	90.2	4.9	53.8	82.3	3.0	94.6	34.9	48.2
Clef-flash (9B)	89.1	4.7	57.4	77.9	2.7	94.1	35.4	52.3
pplx-decider	91.8	4.9	61.5	77.1	2.3	93.9	38.0	51.7
Kev-9B	87.1	4.7	39.0	67.1	2.4	99.5	2.2	83.9
Kev-27B	94.0	4.9	54.8	76.6	2.1	96.7	21.1	66.6
Luna verbal	79.5	3.9	45.0	64.0	2.2	93.9	30.4	62.6
Haiku verbal	74.9	3.4	30.2	65.8	1.7	96.1	21.3	70.7
gpt-oss verbal	83.0	4.4	39.1	0.0	–	–	0.0	100.0
Sonnet verbal	89.9	4.5	52.8	82.7	2.5	94.6	35.8	50.4
prevalence baseline	75.0	3.6	0.0	0.0	–	–	0.0	100.0

Accepted: share of test decisions; error: among them; share 'absent': accepted decisions that say a code does not apply; applying codes found: applying codes accepted with the right label, over all applying codes; deferred: applying codes left for a person. The baseline's naive scores are fitted on all calibration comments; for corpus LTT they are fitted on half and certified on the other half. Design-weighted.

4.2 RQ2: Does uncertainty sit where humans disagree?

Answer: yes, moderately, for most coders. When annotators disagree, a single majority label is a weak target [4, 46], so we ask whether model uncertainty is high where humans split. The evidence has a clear limit: MFRC has 1,382 test comments with three annotators, but AnnoMI's multi-annotated test items (91 client and 93 therapist utterances) come from only four transcripts, so the AnnoMI results describe those conversations.

Uncertainty rises where humans split. For every decision model and the Luna, Haiku, gpt-oss and Sonnet verbal signals, the Spearman correlation between model uncertainty and the entropy of the human labels is 0.24–0.43 on MFRC and 0.33–0.65 on AnnoMI, except Kev-9B on AnnoMI client (0.18) (Figure 4). The signals that rank errors poorly also miss disagreement: token probabilities, vote-based agreement and Qwen3.5-9B's verbalized confidence track it weakly or not at all, and Luna's token-probability uncertainty is unrelated or inversely related to it (-0.10 and -0.03 on AnnoMI). The same signals that rank errors poorly also track human disagreement poorly. Coders run on a subset (Qwen3.8-2.4T, self-consistency) have only 20–23 multi-annotated AnnoMI items and are not interpreted.

No coder falls below the range of individual annotators. In a leave-one-annotator-out comparison, each annotator in turn is scored against the majority of the other annotators, and each coder is scored against the same references (the dot is its mean over references). No coder falls below the range of individual annotators, and some fall above it: on AnnoMI therapist, Jev's mean κ (0.891) exceeds all ten annotators' (0.631–0.886), and on MFRC Sonnet exceeds five of six. This is a descriptive comparison with a broad human range, not a test of equivalence to a typical annotator.

What it means. Low model confidence is partly a signal of a contested item, not only of a likely model error. A triage rule that sends low-confidence items to a person therefore sends them, among others, the items where human judgement matters most. Agreement with annotators, which RQ2 measures, is not the same as interpretive validity, which it does not.

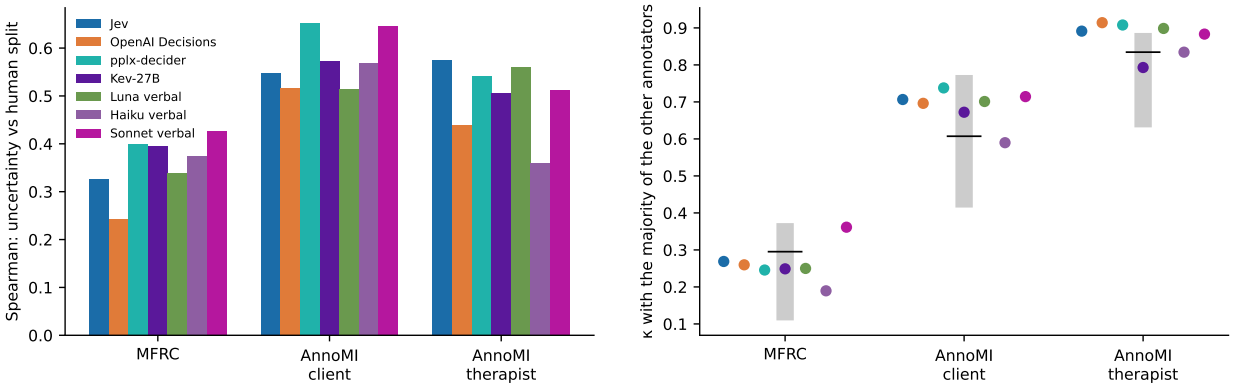


Figure 4: RQ2. Left: Spearman correlation between model uncertainty and human label entropy (MFRC: 1,382 comments; AnnoMI: 91–93 utterances from four transcripts). Right: κ against the majority of the other annotators (grey bar: range over individual annotators; tick: median; dots: each model's mean over the same references). Unweighted.

4.3 RQ3: How stable are decisions?

Answer: mostly stable to rewording, less so to option order, and dependent on context. We perturbed the request on the test split (Figure 5). Jev, OpenAI Decisions, Clef-flash and pplx-decider ran every arm on all applicable tasks; Kev-9B ran one to four tasks per arm, because its hosting stopped; Luna and Sonnet ran the option-order and label arms on 85 test segments per task. Flip rates and changes in AURC, the two pre-specified outcomes, are design-weighted, with per-task cluster-bootstrap intervals (Table 12).

- **Option order matters, differently by model.** Averaged over tasks, reversing or shuffling options changes 4% of Jev's labels, 5–7% of pplx-decider's and Sonnet's, 8–11% of OpenAI Decisions' and Luna's, under 1% of Clef-flash's, and 13–31% of Kev-9B's; in single tasks it reaches 55% (Kev-9B, reversed options, TalkMoves teacher). For coders whose identical repeated requests already vary (2.1% of Jev's labels, 3.3% of Sonnet's, 6.6% of Luna's; §5.6), some order flips may reflect that variation; the samples differ, so the rates cannot simply be subtracted.
- **Wording matters a little.** Paraphrased definitions change 3–10% of labels, and replacing code names with neutral identifiers 5–10%. Asking for MFRC codes one at a time instead of together changes at most 4% (Clef-flash); OpenAI Decisions' multi-label answers are exactly invariant to question order and batching.
- **Removing the previous turn removes information.** It changes 15–24% of dialogue labels on average over tasks (up to 39% in one task) and makes the coders worse: κ falls by 0.012–0.041 and AURC rises by 0.010–0.037. These flips are mostly a loss of relevant context, not instability, in line with the finding that added context can redirect decision models [67].
- **Question format has model-specific effects.** Asking each exclusive option as a separate yes/no question improves error ranking for Clef-flash (+0.029) and pplx-decider (+0.022), hurts it for Kev-27B (−0.028), and produces no clear pooled change for Jev or OpenAI Decisions.

Unstable decisions are mostly uncertain ones. For every coder, the decisions that flip under surface changes are mostly ones it was already unsure of: a stable decision has higher original confidence than a flipped one with probability 0.80 (Luna) to 0.96 (Jev), as the median over each coder's task–arm cells. Changes in AURC under surface perturbations stay within ±0.08 per task for every coder except Kev-9B, whose option-order sensitivity raises AURC by up to 0.23.

What it means. Instability generally concentrates at lower confidence, where triage can catch it. The exceptions matter in practice: the hand-coded sample and the rest of the corpus should be coded with the same request, the same option order and the previous turn, and a model with strong position bias, such as Kev-9B, needs its option order fixed or averaged over; averaging over option rotations [71] is a cheap remedy we did not test.

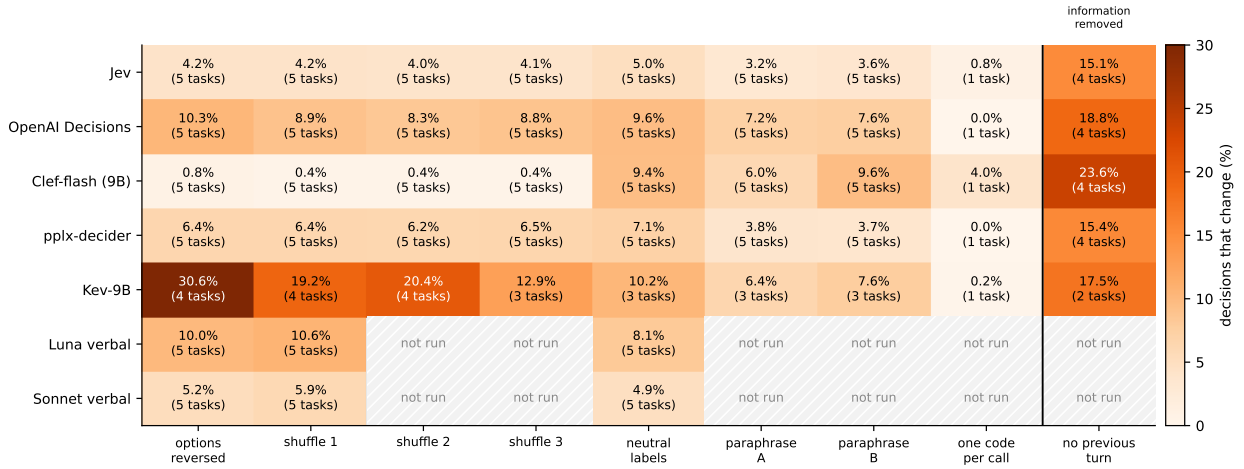


Figure 5: RQ3: share of decisions that change under each perturbation (design-weighted mean over the tasks each cell covers; task counts in brackets). Hatched cells were not run. "No previous turn" removes information rather than rewording the request.

5 Experimental questions: what it takes to triage

The research questions showed what happens; the experimental questions ask why, and what it means in practice. The analyses were specified in amendments v1.1–v1.4, but their details were fixed after the primary results had been seen, and the corpus guarantee was added and revised after external reviews. All are exploratory (Appendix A).

5.1 EQ1: Does calibration error predict triage?

Answer: no. Calibration error is cheap to fix and says little about how well a coder ranks its errors, which is what triage depends on.

Calibration is cheap to fix. Raw calibration errors range from 0.05 to 0.33 (Table 4). A Platt or isotonic map from confidence to accuracy, fitted per task on the calibration split, brings every signal's mean ECE over tasks to 0.03–0.07 (Figure 9 and Table 14, appendix). The fitted Platt map is increasing for every decision model and verbalized signal, so it leaves their ranking unchanged; for the token probabilities and vote shares, which concentrate near 1, the fitted map can flatten or even reverse it. Temperature scaling of the full probability distribution, by contrast, is fragile here: models that output exact zeros and ones make the fit depend on how those are floored before taking logarithms (Table 15). With a floor of 10^{-4}, temperature scaling calibrated Jev well and appeared to lower the token probabilities' error ranking (Luna 0.558 to 0.489); with 10^{-12} it leaves Jev's ECE unchanged and raises Luna's ranking to 0.579. We therefore use Platt maps as the recalibration yardstick and draw no conclusion from temperature scaling.

Low calibration error does not mean good ranking. Clef has one of the lowest raw ECEs (0.055) but ranks errors worse than Jev (ECE 0.120). The raw reliability curves show how the coders are miscalibrated: Jev and OpenAI Decisions are overconfident at high confidence, Kev-27B is underconfident, and Sonnet and pplx-decider are close to calibrated. ECE here is top-label calibration against the majority label; calibration against the distribution of human labels [4] is a different target, which RQ2 approaches.

Raw probabilities mislead the overconfident coders. Taken at face value, "accept if confidence $\geq$0.95" accepts 36–64% of decisions for Jev, OpenAI Decisions, Luna and gpt-oss at 12–14% error, and almost every decision at 24–33% error for the token probabilities and Qwen3.5-9B's verbalized confidence. The other decision models, Haiku, Sonnet and Qwen3.8-2.4T stay below 5% error, mostly because few of their decisions reach 0.95 (4–31%).

What it means. A coder should not be chosen by its calibration error or by a vendor's claim of calibration. If its probabilities will be read as probabilities, recalibrate them on the hand-coded sample; to choose between coders, compare how well they rank errors and how much they accept at the target. A fixed rule such as "accept $\geq$0.95" is unsafe for an overconfident coder.

5.2 EQ2: Do threshold rules keep their target, and what does a guarantee cost?

Answer: the usual rule does not; a guarantee that follows the sampling design does, at a cost. The cost is modest on Reddit comments and prohibitive on dialogue at our budgets.

The naive rule overshoots. Over 500 random cluster re-splits per task, the naive rule's held-out error exceeds 5% in 47–52% of MFRC re-splits for Jev, Clef, Clef-flash, pplx-decider and both Kev models, and in 28% for OpenAI Decisions (Figure 6; per-task rates in Tables 17 and 18). The reason is simple: a threshold chosen to sit just under 5% on calibration data lands on either side of 5% on new data about equally often. The LLMs' verbalized confidence overshoots less on MFRC (0–19%), probably because its coarse confidence levels put the threshold at a tie boundary with room to spare. On the dialogue tasks, wherever a decision model's naive rule accepts more than 5% of decisions, its held-out error exceeds 5% in 37–85% of re-splits.

A guarantee that ignores clustering fails. LTT on decisions treats the eight codes of a comment, and the utterances of a transcript, as independent. They are not: a model that misreads a comment or a stretch of conversation can get several of its decisions wrong together. On MFRC this rule's held-out error exceeds 5% in 5.8–11.2% of re-splits for every decision model, above the nominal 5%. In simulations with calibrated confidence and errors correlated within clusters (Table 20), the population risk of its threshold exceeds 5% in 6% of runs with independent errors, in 14% with strong correlation in MFRC-like clusters of eight, and in 18–31% with dialogue-like clusters; the error on the uncoded remainder exceeds 5% in 10–30% of runs.

The corpus guarantee holds on comments. Corpus LTT (§3.4) never exceeds 5% held-out error in any re-split of any task. Because each test split is the uncoded remainder of a random split, this is a direct check of its guarantee. LTT on i.i.d. clusters never exceeds it either, and in simulation both exceed 5% in at most 0.3% of runs, for either target. On the frozen MFRC split, corpus LTT certifies 60–83% of decisions for six of the seven decision models and for Luna's, Haiku's and Sonnet's verbalized confidence, at realized test errors of 1.6–3.0% (Table 6). It certifies nothing for OpenAI Decisions (at its highest thresholds its calibration error, 2.5–3.8%, is too close to 5% for a test corrected for 16 thresholds), gpt-oss, Qwen3.8-2.4T (its subset has too few calibration comments), the token probabilities or the agreement signals. The frozen split is one draw: over the re-splits, corpus LTT accepts on average 65–85% of MFRC decisions for the same coders, and 64% for gpt-oss.

With whole transcripts, dialogue cannot be certified. With whole transcripts or lessons hand-coded, as in our splits, corpus LTT certifies nothing on the dialogue tasks, for any coder, and with this procedure at our calibration sizes nothing better was possible. The reason is the sampling unit. The test caps its bets and scales each unit by B, the weight of the largest unit, so an error-free unit multiplies the evidence by at most $1 + 0.05 \times$ (its weight$/B$), whatever the order, and the evidence must reach $16/0.05 = 320$ (Appendix B). A typical transcript carries only 8–27% of the largest one's weight, so each adds little. On AnnoMI and TalkMoves student, even a coder that is never wrong could not be certified, whichever transcripts were hand-coded; on TalkMoves teacher, fewer than 388 of the 444 lessons cannot suffice (Table 21).

Sampling utterances removes that obstacle, but the evidence is still too thin. The corpus guarantee allows any dependence, so single utterances drawn at random, each coded with its context, are valid units too. They weigh about the same, and a perfect coder is then certified after a median of 139–163 hand-coded segments. Under that design, with 30% of segments hand-coded, the real coders still certify almost nothing (Table 22): pplx-decider and Kev-27B on AnnoMI therapist obtain a certificate in 2 and 1 of 200 re-splits (accepting about 43% and 39% of decisions when they do), and the other coder–task pairs in none. The confident accepted sets supply too little evidence of $\leq$5% error. On the first re-split's roughly 430

hand-coded segments, 11 of the 44 coder–task pairs have no grid threshold at or below 5% error against the majority label; 30 have one, but it accepts at most 65 segments (median 9); and the three larger qualifying sets (191–227 segments, on AnnoMI therapist) sit at about 4.5%, too close to 5% for a conservative test to separate. On the same re-splits the naive rule exceeds 5% held-out error in up to 78% of them. On dialogue, then, the binding constraints are the sampling design first and then the amount and accuracy of confident evidence, not dependence among utterances. LTT on decisions appeared to certify 42% (Kev-27B) and 24% (Sonnet) on AnnoMI therapist under the transcript design; those certificates were not valid.

How much hand-coding a guarantee needs. The calibration-budget curves (Figure 10, appendix) draw smaller calibration sets of whole MFRC comments, keeping the test split as the remainder. Corpus LTT certifies nothing with 50, 100 or 200 hand-coded comments, and 2% (Jev) to 78% (Sonnet) with 400. The naive rule accepts as much with 50 comments as with 600, but exceeds 5% held-out error in up to 54% of draws (pplx-decider) and still in 27% with 400. Testing one threshold fixed in advance (0.95) instead of the grid halves the comments a perfect coder needs (64). It certifies gpt-oss (72%) but still not OpenAI Decisions, and it raises Luna's share (64% to 75%); for the other certified coders the fixed threshold is restrictive and certifies less than the grid, for example Jev 17% instead of 60% (Table 19).

Thresholds do not transfer. A threshold fitted on the other four tasks exceeds 5% error in 52 of the 64 task–coder cells where it accepts anything (median 16% on the dialogue tasks), and a fixed "accept $\geq$0.95" rule in 52 of 75. Guarantees do not travel either. Corpus LTT does not apply, because the other tasks are not a sample of the new corpus, and LTT on i.i.d. clusters pooled over the other four tasks certifies nothing at all. This matches Lu and Wu [37], who find that rejection thresholds of typed decision models do not carry across datasets.

What it means. If the error target must hold, a researcher should hand-code a random sample from their own corpus and use a rule whose unit matches how that sample was drawn. With our 16-threshold grid, even in the best case nothing is certified until about 120 error-free units of similar weight have been hand-coded, so small units (single comments or utterances, or fixed-size blocks) are far cheaper than whole transcripts of very unequal length. Thresholds borrowed from another codebook, or fixed in advance at a round number, should not be trusted.

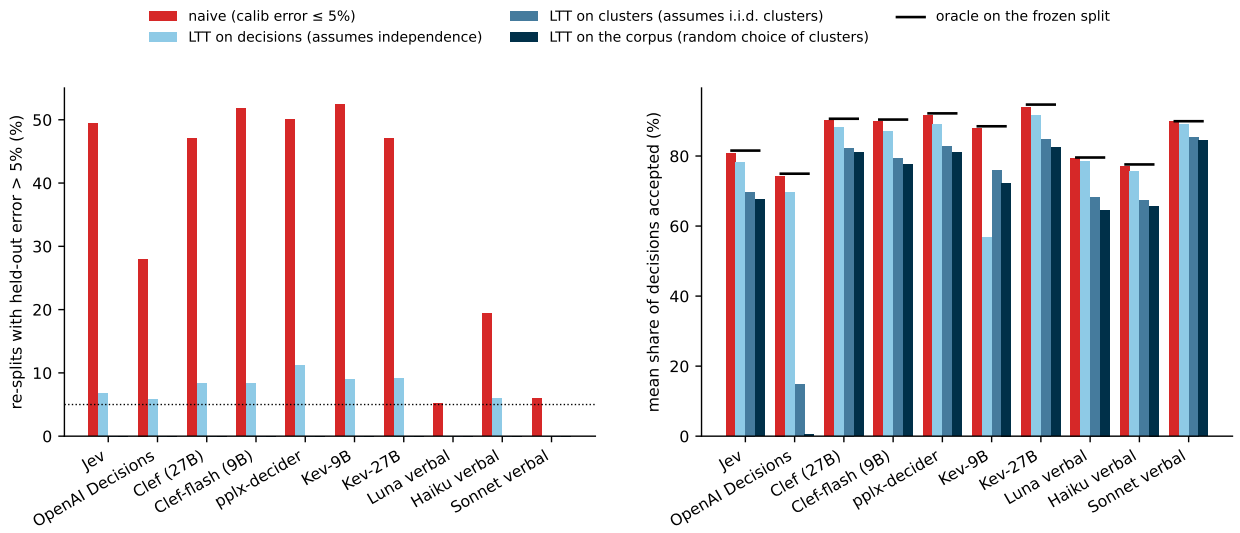


Figure 6: EQ2 on MFRC, the only task where the guaranteed rules accept anything when whole clusters are hand-coded, over 500 random cluster re-splits. Left: share of re-splits whose held-out error exceeds 5% (dotted: 5%). Right: mean share of test decisions accepted; black ticks mark the oracle on the frozen split. LTT on decisions assumes independent decisions and LTT on i.i.d. clusters independent clusters; corpus LTT assumes only that the hand-coded clusters are a random sample, visited in random orders.

5.3 EQ3: Ranking or ties at the top?

Answer: both. For Jev and OpenAI Decisions, large blocks of decisions at the very top of the confidence scale, with too many errors in them, limit how much can be accepted. For every coder with enough data, recalibrating per code improves ranking, partly by using which code was predicted.

Large, error-prone ties at the top. Even a threshold fitted on the test set itself (the oracle) accepts only 16% of Jev's decisions at $\leq$5% error, against 40% for Sonnet. Ties at the top contribute to this (Figure 7). On the dialogue tasks Jev places 8–30% of its decisions at exactly its top confidence (1.00), and 6.5–12.3% of those are wrong; OpenAI Decisions behaves the same way (up to 25% of decisions, with up to 12% error). A threshold must accept or reject such a tie as a whole, so for Jev none of the dialogue thresholds reaches 5% empirical error. No other coder puts more than 7.5% of decisions in its top block, and Clef, Kev and pplx-decider put almost none there.

Per-code maps split the ties and narrow the differences. Per-code recalibration fits a separate Platt map for each predicted code (for MFRC, each code) on the calibration split. It splits the ties and improves error ranking for every coder with enough calibration data: by +0.025 (Sonnet, Kev-27B) to +0.12 (Kev-9B, Qwen3.5-9B), and by +0.040 [0.027, 0.054] for Jev (Figure 11, appendix; intervals conditional on the fitted maps). For coders that return exact zeros and ones, the size of the gain depends on the floor applied before the logit: Jev's is +0.053 with a floor of 10^{-4} instead of 10^{-12}, and the token probabilities move by up to 0.024, while the other decision models and verbalized signals change by at most 0.003 (Table 15). Per-code maps also narrow the differences between decision models: six of the seven land within 0.80–0.83 (OpenAI Decisions 0.77), against 0.69–0.79 raw.

Part of the gain is which code was predicted. Scoring each decision only by the calibration accuracy of its predicted code, with no confidence at all, gives 0.66–0.71 (Table 25); for Kev-9B this alone beats its raw confidence (0.713 vs 0.694). Some codes are simply harder than others, and a per-code map learns that. The maps do not make coders equivalent: oracle coverage with them still ranges from 23% (Kev-9B) to 46% (pplx-decider). On identical items, Jev with per-code maps comes close to Sonnet's raw confidence, but the

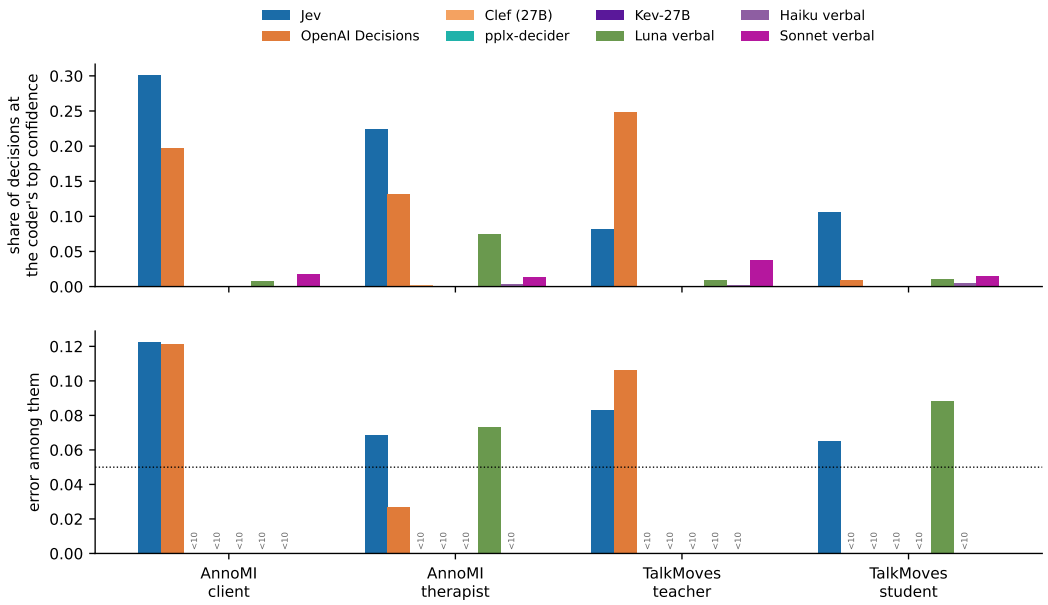


Figure 7: EQ3: top-confidence tie blocks on the dialogue tasks. Top: share of decisions at the coder's maximum confidence. Bottom: error among them; blocks of fewer than 10 decisions are omitted. Above the dotted line, the top block itself exceeds the 5% target.

interval does not resolve the difference either way (-0.004 [-0.018, 0.008]), and it trails Sonnet with its own per-code maps (-0.030 [-0.041, -0.019]; significant on MFRC, AnnoMI therapist and TalkMoves student).

Better ranking did not buy certified coverage at our calibration sizes. A learned map must be fitted on clusters that are not then used to certify it, so the maps were fitted on half of the MFRC calibration comments and certified on the other half (304 comments). Corpus LTT then certifies 66–75% for Clef, Clef-flash, pplx-decider and Sonnet, against 77–83% with their raw confidence on all 608 comments, and nothing for the other coders.

What it means. Per-code maps are worth fitting, since they improved ranking for every coder with enough data, but they cost hand-coded data. They must be fitted on a pilot sample kept apart from the sample used to certify, and with a fixed budget the better ranking can be outweighed by the smaller certification sample, so both routes should be compared under the same budget.

5.4 EQ4: Do related 9B decision models outperform a 9B instruct model's confidence?

Answer: against its verbalized confidence, yes; against its token probabilities, barely. Clef-flash and Kev-9B are built on Qwen3.5-9B (Kev-9B on the base checkpoint; Clef-flash on a variant Cloudflare does not specify), and we ran the Qwen3.5-9B instruct model as a plain LLM (Table 8; pre-specified secondary). These systems differ in training, read-out and request format, so this compares related systems, not a controlled ablation.

Most of the gain is over verbalized confidence, little over token probabilities. Both decision models rank errors better than the instruct model's verbalized confidence (+0.065 and +0.048 AUROC). Against its answer-token probabilities the gain is small: +0.027 [0.004, 0.052] for Clef-flash, and not significant for Kev-9B. Under the naive 5%-target rule both accept 15–22 points more, consistent with their less concentrated scores. Clef-flash and Kev-9B do not differ significantly.

Table 8: EQ4: two 9B decision models built on Qwen3.5-9B vs the Qwen3.5-9B instruct model's own confidence (pooled over tasks).

comparison	metric	difference	95% CI	p (bootstrap)
Clef-flash (9B) – Qwen3.5-9B verbal	error AUROC	0.065	[0.048, 0.085]	<0.001
Clef-flash (9B) – Qwen3.5-9B verbal	coverage@5%	0.221	[0.181, 0.297]	<0.001
Clef-flash (9B) – Qwen3.5-9B logprob	error AUROC	0.027	[0.004, 0.052]	0.026
Clef-flash (9B) – Qwen3.5-9B logprob	coverage@5%	0.161	[0.110, 0.211]	<0.001
Kev-9B – Qwen3.5-9B verbal	error AUROC	0.048	[0.030, 0.066]	<0.001
Kev-9B – Qwen3.5-9B verbal	coverage@5%	0.205	[0.152, 0.280]	<0.001
Kev-9B – Qwen3.5-9B logprob	error AUROC	0.010	[-0.018, 0.036]	0.512
Kev-9B – Qwen3.5-9B logprob	coverage@5%	0.146	[0.079, 0.189]	<0.001
Clef-flash (9B) – Kev-9B	error AUROC	0.018	[-0.007, 0.043]	0.142
Clef-flash (9B) – Kev-9B	coverage@5%	0.015	[-0.043, 0.097]	0.560

Coverage is the naive rule's share accepted. Kev-9B starts from the base checkpoint and Clef-flash's exact variant is unspecified, so these contrasts compare related systems with different training and read-out, not a controlled training ablation.

Why the read-out may matter. Our token probabilities are read after a one-sentence rationale, which may commit the model and push its token probabilities towards 1. Li and Wagle [32] read next-token probabilities over option identifiers directly, without generated text, and find that a 4B base model matches decision models built on the same backbone. Together these results suggest, as a hypothesis, that much of a decision model's advantage comes from reading out a probability rather than from its training.

What it means. A researcher with a capable open model may get much of a decision model's benefit by reading the probabilities of the option labels directly, without a generated rationale. We did not test this; a matched-checkpoint experiment with direct option scoring would.

5.5 EQ5: Which configurations give the most guaranteed automation per dollar?

Answer: on comments, pplx-decider gives the most certified automation per inference dollar, and several decision-model configurations come within two points of Sonnet at a fraction of its cost. But nearly all certified decisions are "absent" calls, and a person still reads most comments.

What each configuration certifies, and at what cost. When whole clusters are hand-coded, as in our splits, certificates exist only on MFRC, so we compare configurations there (Figure 8), each at the API cost of its own policy. Table 27 gives the certified results and Table 26 pooled results under the naive rule and the oracle.

- **A coder that reads nothing is not certified.** The prevalence baseline accepts 75% of MFRC decisions at 3.6% error under the naive rule, all of them "absent". Its scores are learned from labels, so they must be fitted on half of the calibration comments and certified on the other half, and corpus LTT then certifies none of its decisions.
- **Single decision models:** pplx-decider certifies 77% at $0.03 per 1,000 comments, Clef-flash 78% at $0.09 and Clef 82% at $0.24; Sonnet certifies 83% at $4.52.
- **A pplx-decider + Kev-27B agreement gate**, which accepts where both give the same label, certifies 81% at $0.34. Its difference from Sonnet is −1.6 [−10.0, +3.8] points: not significant, which is not the same as equivalent.
- **Kev-27B** certifies 77% at an estimated $0.32 per 1,000 comments, significantly more than Jev's 60% (+16.7 [+7.1, +38.2] points).
- **Chains are certified as a whole** (§3.4), and the larger joint test costs power: Kev-27B → Sonnet certifies 82% and Jev → Sonnet 77%, no more than Sonnet alone.
- **The seven-model committee** certifies 84%, slightly more than its best member (Clef, 82%), at four times Clef's cost ($0.97); its difference from Sonnet is +1.8 [−8.3, +2.8] points.

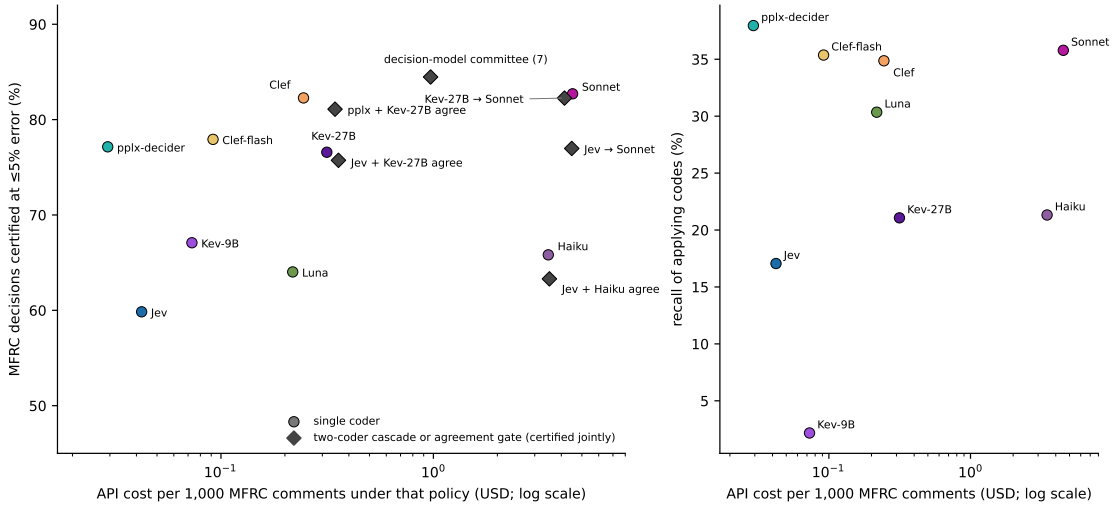


Figure 8: EQ5 on MFRC. Left: share of decisions certified at ≤5% error by corpus LTT vs API cost per 1,000 comments under the same policy (log scale); diamonds are two-coder configurations, certified jointly. Right: share of the codes that apply that the certified decisions label correctly. Coders that certify nothing (OpenAI Decisions, gpt-oss, the token probabilities, the agreement signals and the prevalence baseline) are not shown. Kev's cost is an estimated GPU serving cost.

Accepted decisions are not reading saved, or codes found. A comment leaves the human queue only when all eight of its codes are accepted: under Sonnet's certified policy, 66% of the uncoded remainder's comments still need a person to read at least one code, and under Jev's 93%. Counting the 608 hand-coded calibration comments as well, a person reads 76% and 95% of all comments (Table 27). And 94–99.5% of the certified decisions say that a code does not apply: pplx-decider's certified decisions label 38% of the codes that apply, Sonnet's 36%, Jev's 17% and Kev-9B's 2% (Figure 8, right; Table 7).

Inference cost is not the whole cost. Costs are list prices for inference only. They exclude the hand-coded calibration sample, which a guarantee requires in any case, and the human reading that remains; Kev's figure is an estimated warm-serving cost. Human reading can dominate: per 1,000 remainder comments, Sonnet costs about $4.49 more than pplx-decider in inference and leaves about 75 fewer comments for a person. Assuming the same cost per human-read comment under both policies, Sonnet has the lower combined inference-plus-reading cost on the remainder once that cost exceeds about $0.06; this is a sensitivity calculation, not a measured reading time.

What it means. At these prices inference costs at most about $8 per 1,000 comments, so the choice between coders should rest on how much reading each saves and how many applying codes it finds, not only on its accepted share or its price. Once reading is priced in, a more accurate but costlier coder can be the cheaper policy overall.

5.6 EQ6: Are the results artefacts?

Answer: the checks bound the main threats rather than proving validity, and one of them suggests errors in some human labels. Table 9 summarizes four checks.

- **Repeatability.** In the repeated requests we tested (100 test segments per task, sent twice), OpenAI Decisions, Clef, Clef-flash and pplx-decider returned identical outputs, and Haiku changed 0.03% of labels. Jev changed 2.1%, almost all at low confidence; Sonnet, which runs at the provider's default sampling settings, 3.3%; and Luna 6.6% despite temperature 0.
- **Prompt sensitivity** (v1.4). One alternative prompt, which asks for a probability for every option [54], does not improve the verbalized baselines: Sonnet −0.034 [−0.056, −0.011], Haiku −0.010 [−0.045, 0.022].

Table 9: EQ6: validity checks. Run-to-run flips: identical requests sent twice (100 test segments per task; 0 means no observed label changes in these repeats, <0.001 a nonzero rate below display precision); option-order flips: reversed and shuffled options (RQ3 arms); memorization probe with decoy-name control (amendment v1.3, deviation 14). Decision models return no text, so they could not be probed.

coder	run-to-run label flips	flips under option order	completion-probe flag	label recall beyond decoy
Jev	0.021	0.042	not assessed	not assessed
OpenAI Decisions	0	0.096	not assessed	not assessed
Clef (27B)	0	–	not assessed	not assessed
Clef-flash (9B)	0	0.006	not assessed	not assessed
pplx-decider	0	0.064	not assessed	not assessed
Kev-9B	–	0.249	not assessed	not assessed
Kev-27B	–	–	not assessed	not assessed
Luna verbal	0.066	0.103	not flagged	MFRC
Haiku verbal	<0.001	–	not flagged	no evidence
Sonnet verbal	0.033	0.056	not flagged	no evidence
gpt-oss verbal	–	–	not flagged	no evidence
Qwen3.5-9B verbal	–	–	not flagged	no evidence
Qwen3.8-2.4T verbal	–	–	not flagged	no evidence

Completion-probe flag: the pre-specified rule (Table 33); not flagged does not mean no literal matches, which occurred rarely under both prompts. Prompt sensitivity (amendment v1.4): a verbalized-distribution prompt changes error AUROC by -0.034 [-0.056, -0.011] for Sonnet and -0.010 [-0.045, +0.022] for Haiku.

Our canonical prompt does not appear to handicap them, though one alternative cannot rule out a better one.

- **Memorization** (v1.3). Given the first half of 100 items per dataset, no model–dataset pair met the pre-specified completion flag, and naming the dataset never significantly increased overlap with the true continuation. Near-verbatim matches were rare and occurred with and without the name; they include a predictable counting sequence in a TalkMoves item and Du Bois quotations in Dunivin passages (Table 32). Naming the dataset raised label recall in three model–task cells; a decoy name with the same description explains two as task framing, and the third (Luna on MFRC, +0.13 over the decoy) does not raise Luna's coding accuracy on those items. Decision models return no text and could not be probed.
- **Label noise and replication.** When Jev, Luna, Haiku or gpt-oss disagreed with the human label on a sampled, high-confidence-enriched item, adjudication sided with the coder in 55–64% of cases (Table 31); counting only the author's verdicts (the disputed items the author adjudicated and the 12-item audit of panel-agreed items), it did so in 52–62%. This echoes Liu et al. [36], but the sample cannot estimate the overall rate of label error. Our harness recomputes the agreement Dunivin reports for his GPT-4 runs from his released predictions [12], exactly, and Sonnet's mean κ on his test set (0.679) is close to his best reported configuration (0.68).

What it means. Some coders change a few labels when the same request is sent again, so the model version and its outputs should be recorded with the coded data. And because some majority labels appear to be wrong, part of a measured "model error" may be a disagreement the model is right about; Section 6 returns to what this means for a 5% target.

6 Discussion

Whether model confidence can triage qualitative coding turns out to depend less on which model is used than on three choices a researcher makes: how the confidence is judged, how the hand-coded sample is drawn, and how automation is counted. We take each in turn, add what the results say about the human labels themselves, and then place the findings in prior work and draw out what they mean for researchers and for the builders of decision models.

Judge confidence by its ranking, on your own data. Calibration error was the wrong yardstick throughout. It is cheap to fix after the fact (§5.1), and it says little about what triage needs, which is that a coder's mistakes receive lower confidence than its correct labels. That ranking varied widely within the decision-model category, from worse than a cheap LLM's verbalized confidence to close to a frontier LLM's (§4.1), and it varied across tasks for the same coder. Two mechanisms appear to be at work. Large ties at the top of the scale, which no threshold can split, limit some coders (§5.3). And the read-out may matter: two decision models rank errors clearly better than the verbalized confidence of a related instruct model, though only slightly better than its token probabilities, which is consistent with the hypothesis that reading out a probability is much of the advantage (§5.4). Per-code recalibration narrows most of the remaining differences, partly by learning which codes are hard. None of this can be read off a product description; it has to be measured on the researcher's own codebook.

A guarantee is a property of the sampling design. The usual practice, tuning a threshold to 5% error on hand-coded data, misses its target about half the time, and a standard guarantee that treats decisions as independent fails because decisions within a comment or a transcript err together (§5.2). What made a guarantee valid was not a better model but a better account of how the hand-coded sample was drawn: our corpus rule needs only that the hand-coded units are a random sample of the corpus. The same account explains its cost. Whole transcripts of very unequal length carry little evidence each, so on dialogue nothing could be certified at our budgets. Sampling single utterances removed that obstacle, although the coders' confident dialogue decisions then gave too little evidence. Hand-coding for a guarantee is therefore a sampling-design decision, of the kind survey methodology has long studied [24], and it should be planned before coding starts.

Count reading saved and codes found, not decisions accepted. In a sparse multi-label codebook most decisions are easy calls that a code does not apply. A coder can accept most of them, and a baseline that reads no text accepts 75% at low error, while finding few of the codes that apply and leaving most comments for a person to read (§4.1, §5.5). Accepted share is a convenient number, but the quantities a researcher feels are the codes found and the segments still read. On those, the differences between coders are larger than on accepted share, and a more accurate, costlier coder can save more work than a cheaper one.

Low confidence partly marks contested items. Model uncertainty was higher where human coders disagreed (§4.2), no coder fell below the range of individual annotators, and adjudication often sided with the model on sampled disagreements (§5.6). Error measured against a single majority label may therefore overstate some model errors, and a 5% target against such a reference is partly a target against annotator noise [46, 58]. Triage then has a useful side effect: the items it sends to a person include many on which human judgement is genuinely needed. Agreement with annotators, which we measure, is still not the same as interpretive validity, which we do not.

Relation to prior work. Where our design overlaps with the concurrent evaluations of decision models [25, 51], the findings agree: Jev trails the strongest LLMs in accuracy, its raw calibration beats most LLM signals but not the frontier models, and cascades save cost. Our results add a caution to coverage reported at fixed confidence thresholds, there and in confidence-based routing of qualitative codes [72]: such thresholds need to be checked on held-out data from the same corpus, or certified, because they overshoot and do not transfer. That they do not transfer matches Lu and Wu [37]; the read-out hypothesis is consistent with Li and Wagle [32]; and the evidence of errors in some majority labels echoes Liu et al. [36]. Guarantees for annotation already exist [70]; our contribution there is to show that, for coding data, which guarantee is valid depends on how the hand-coded sample is drawn.

A recipe for model-assisted coding. For a researcher who wants to code a corpus with a fixed codebook and check only part of it by hand, the results suggest four steps.

1. Hand-code a random sample of units from the corpus itself, and set part of it aside for certification. Units can be comments, transcripts, random fixed-size blocks of turns, or single utterances coded with their

context; units of similar weight make a guarantee far cheaper (§5.2). Random selection is what makes the guarantee possible; thresholds from other codebooks, and fixed rules such as "accept ≥0.95", did not carry over in our data.

2. On the other part, a pilot sample, choose the coder and its scores. Compare how many decisions each candidate accepts at the target, which after recalibration depends mostly on accuracy; low ECE is not a sufficient criterion (§5.1). Fit any per-code maps here: they improved error ranking for every coder with enough calibration data, but maps fitted on the certification sample would void its guarantee (§5.3). Compare the accepted share with a prevalence baseline that reads no text, and check how many applying codes the accepted decisions contain (§4.1).
3. Freeze the chosen coder, scores and threshold grid, then certify on the set-aside sample with a rule that respects how it was drawn, such as corpus LTT. Each certificate covers one configuration: choosing among configurations by their certification results needs all of them in one testing family. With our 16-threshold grid, even in the best case it needed about 120 error-free units of similar weight before it certified anything; with whole transcripts of very unequal length it could not certify even a perfect coder (§5.2).
4. Count what a person still has to read, including the hand-coded sample: in multi-label coding, a segment leaves the queue only when all its codes are accepted (§5.5).

For builders of decision models. Decision models are marketed as calibrated, and most of the seven are close to it, or can cheaply be made so. The results point to what would make them more useful for coding: reporting error ranking and risk–coverage on human-coded data, not only calibration; resolving the top of the confidence scale instead of placing many decisions at exactly 1.0 (§5.3); making outputs invariant to option order, or averaging over orders, since order changed up to 55% of one model's labels in a single task (§4.3); and documenting which outputs are deterministic (§5.6).

From labels to estimates. Triage bounds the error rate among the labels a researcher accepts, in aggregate; it says nothing about any one label. Many qualitative studies go on to report code frequencies or associations between codes. Prediction-powered and design-based estimators correct such quantities using a hand-coded subsample, without per-item guarantees [2, 15], and model confidence can decide which items to hand-code [18]. The calibration sample that triage needs can serve these estimators too, and they do not need a certified threshold, which makes them attractive on dialogue data, where we could certify almost nothing under either sampling design.

7 Limitations and follow-ups

Each limitation below comes with what would address it.

- **Scope.** Five tasks on short segments (comments, utterances, sentences) from three public datasets, and pooled estimates describe these tasks, not a population of codebooks. We have no long-form interviews: public interview data with applied codes is rare, and a private dataset requires ethics approval. A coded interview corpus, used with consent for third-party processing, would test whether the results carry over to the setting where qualitative coding is most common.
- **A dated snapshot.** Several APIs were in public beta, and the models will change. We release the harness, the analysis plan and every coding decision, so every number can be recomputed from the public datasets and the study re-run on newer models; the raw response text is withheld because it quotes the datasets.
- **Guarantees.** Corpus LTT assumes that the hand-coded clusters are a uniformly random subset of the corpus and that the test visits them in random orders; our split assigns clusters by a hash of their id, which we treat as random, and the orders come from a recorded seed. Each certificate covers one configuration. The procedure is conservative (in simulation it accepted less than the best safe threshold), and much of its cost on dialogue comes from scaling every transcript by the largest one. Sampling single utterances avoids most of that, although at our budget the coders' confident dialogue decisions still gave too little evidence. Procedures that handle unequal unit weights more efficiently, or larger hand-coded samples of small units, could certify more. The population version additionally needs independent clusters and a

true bound on cluster weight; for it, random re-splits are only a diagnostic, while for the corpus guarantee they are a direct check.

- **Post-hoc analysis.** The plan was kept in the project repository, not externally registered. Sonnet was extended to the full sample after subset results were seen, the A3 analyses were revised after their first output, and the corpus guarantee, the prevalence analysis and several controls were added or revised after external reviews. All are labelled exploratory; a registered replication on new codebooks would confirm or correct them.
- **LLM read-out.** Token probabilities were read after a generated rationale, and multi-label verbalized probabilities were converted under a reading the prompt did not fix (§3); both may disadvantage the LLM baselines. We tested one alternative prompt; reading option probabilities directly, without a rationale, would test the read-out hypothesis of §5.4.
- **Adjudication** covered a stratified, high-confidence-enriched sample of 200 items drawn from disagreements with six label sources. It relied partly on an LLM panel whose families overlap with the coders it judged (Claude Opus with Haiku and Sonnet; GPT-6 Sol with Luna and OpenAI Decisions), and the author, the only human adjudicator, did not repeat a sitting to measure their own consistency. Independent expert adjudicators would make the label-noise estimate firmer.
- **Measurement.** RQ2 rests on four AnnoMI transcripts and is unweighted. ECE is top-label calibration against the majority label. Costs are October 2026 list prices for inference only, Kev's are estimated warm-serving costs, and human reading is counted in segments, not timed; a timing study would turn the reading counts into hours.
- **Coverage gaps.** Kev-9B's robustness arms cover one to four tasks, and Kev-27B was run on the main and format arms only. Decision models return no text and could not be probed for memorization; Kev's published training data excludes our datasets.

8 Conclusion

Model confidence can triage qualitative coding, but its value is easy to overstate. What matters is how well a coder's confidence ranks its own errors, which calibration error does not reveal and which has to be measured on the researcher's own codebook. An error target holds only under a rule that follows how the hand-coded sample was drawn: with one, models certified most decisions on Reddit comments, almost all of them calls that a code does not apply, and almost none on dialogue at our budgets. Decision models make confidence cheap to obtain, and the best configurations came within two points of a frontier LLM's certified automation on comments at a fraction of its cost. Whether that automation saves real work depends on what is left for a person to read.

Data and code availability

The harness, the analysis plan with its amendments and deviations log, every coding decision (one row per decision, with the probability and the model version returned), the item-level scores of the memorization probe and its decoy control, the samples and splits, the adjudication verdicts, a text-free spend ledger (405,339 requests; $78.56 in API fees, plus GPU time for Kev) and the paper source are at `https://github.com/firish/margin`. The raw text of model requests and responses is not released, because it quotes the datasets; the probe's scores therefore cannot be re-derived from text. The datasets are not redistributed; they are available from their authors under their own terms (MFRC: CC BY 4.0; TalkMoves: CC BY-NC-SA 4.0; AnnoMI and Dunivin: no licence stated).

Ethics statement

All data are public and were collected and annotated by their original authors; we used them only for evaluation. Only these public texts were sent to commercial APIs; no private or newly collected data were used. AnnoMI's counselling conversations are transcribed from publicly available demonstration videos of motivational interviewing. The only human participant was the author, as adjudicator.

Use of AI assistants

The author used an AI coding assistant (Claude) to implement the harness and to draft and edit the manuscript, under the author's direction. An external AI reviewer (GPT-6.1) was used to find errors and loose ends in successive drafts; the author filtered its suggestions and chose which to adopt, and the resulting changes are logged as deviations 16–18. The author takes full responsibility for the content.

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Guide to the appendices

Appendix A records the analysis plan, its amendments and the deviations log. Appendix B defines the guaranteed threshold rules and proves the proposition of Section 3.4. Appendix C collects the supporting results, grouped by the question they support; each group opens with what was done, why, and what it shows. RQ1 and EQ4 are reported in full in the main text. Appendix D shows the requests sent to the models, and Appendix E describes the memorization probe.

A Analysis plan, amendments and deviations

The analysis plan (v1.0) fixed the tasks, splits, decision units, metrics, primary rule and contrasts before any test-split outcome was computed. Four amendments were each recorded before the data they govern. The plan was kept in the project repository rather than registered externally:

- **v1.1** added OpenAI Decisions as a confirmatory coder, Qwen3.5-9B as a non-reasoning token-probability comparator, and the A3 analyses (recalibration, per-code thresholds, risk control, cascades).
- **v1.2** added Clef, Clef-flash, pplx-decider, Kev-9B and Kev-27B as a confirmatory family, with the same-base contrasts pre-specified as secondary.
- **v1.3** added the memorization probe.
- **v1.4** added the prompt-sensitivity check.

The deviations log records eighteen entries; everything added or changed after results were seen is exploratory. In summary:

1. A response-cache key ignored option order. It was fixed and the affected order arms were re-run before analysis.
2. Adjudication moved from a single human pass to a blinded two-family LLM panel plus the author on disputed items.
3. The author completed 48 of 72 assigned adjudications.
4. The RQ5 code-type labels were assigned by the assistant from definition text, after some per-code results had been seen.
5. OpenAI Decisions collection notes, including its exact invariance to question order on multi-label tasks.
6. The Qwen model was chosen by availability without third-party data sharing; Qwen3.8-2.4T was added on a subset; a null-logprob parse fix was made before any full-run data.
7. The A3 implementation details were fixed after the primary results had been seen.
8. A3 was revised after its first output: a power-based LTT start, oracle thresholds and a restructured per-code head-to-head. All A3 results are therefore exploratory.
9. Sonnet was extended from a nested subset to the full sample after the subset results had been seen.
10. The repeatability check, Sonnet's robustness arms and the Holm sensitivity analysis were added.
11. Kev-27B was limited to the main and format arms.
12. Kev-9B's robustness arms cover one to four tasks per arm: the hosting workspace was disabled partway through.
13. The decision-model committee and new cascades were added, exploratory, and fixed before computation.
14. A decoy-name control was added for flagged memorization cells.
15. The author later reviewed those labels and approved them unchanged, after the RQ5 results had been computed.
16. After an external review of the first draft: a cluster-level guarantee replaced LTT on decisions, with joint certification for cascades and split samples for learned scores; the prevalence analysis, the category-only control, adjudication by verdict source, the feasibility calculation and a text-free spend ledger were added; RQ3 point estimates were design-weighted; three reporting errors were fixed.
17. After a second review: the guarantee was recast for a fixed corpus (corpus LTT), with the earlier test kept as LTT on i.i.d. clusters; related tasks share bootstrap draws; the Sonnet subset contrast is the confirmatory one; the probability floor moved to 10^{-12}, and Platt maps became the recalibration yardstick.
18. After a third review: the betting tests average over random visiting orders; the budget curves, bootstrap streams and prompt-sensitivity intervals were corrected; the feasibility bound and the single-utterance design were added; prose intervals are generated from the results files.

The full log, plan and amendments are in the released repository.

B Guaranteed threshold rules

LTT on decisions. Thresholds are tested from the most confident tie block downwards with binomial tests of "risk $> \varepsilon$" at level $\delta = 0.05$, stopping at the first failure; the sequence starts at the first block with at least 124 decisions. Errors are scaled to Kish's effective sample size [28], which gives fractional counts. The test treats decisions as independent and fails when they are not (Table 20).

Corpus LTT. A corpus has N clusters; n of them, chosen uniformly at random, are hand-coded. Fix the scores, the grid $\{0.5, 0.55, \ldots, 0.9, 0.925, 0.95, 0.97, 0.98, 0.99, 0.995, 1.0\}$ and B, the largest total weight of one cluster in the corpus. For threshold t, every cluster has a fixed value $Y_i = (Z_i + \varepsilon B)/B \in [0, 1]$ with $Z_i = \sum_j w_{ij} \mathbf{1}\{c_{ij} \geq t\}(e_{ij} - \varepsilon)$; let μ be their corpus mean and $\bar{Y}$ the mean over the hand-coded clusters. The uncoded remainder has error at most ε among its accepted decisions exactly when its Z-sum is at most 0, that is, when $\mu \leq m^* = ((N - n)\varepsilon + n\bar{Y})/N$.

To test "$\mu \geq m$", visit the hand-coded clusters in a uniformly random order, drawn independently of the labels; the clusters visited are then a uniformly random sample drawn without replacement, in draw order. Under that null, the kth cluster's value has conditional mean $(N\mu - S_{k-1})/(N - k + 1) \geq \mu_k(m) = (Nm - S_{k-1})/(N - k + 1)$, where S_{k-1} is the sum of the values already seen. So $K_k = \prod_{i \leq k}\big(1 - \nu_i\,(Y_i - \mathrm{clip}(\mu_i(m), 0, 1))\big)$, with predictable bets $\nu_i = \min\{1, \sqrt{2\log(1/\delta')/(n\hat{\sigma}^2_{i-1})}\} \in [0, 1]$, is a non-negative supermartingale [62]. The variance estimate is the plug-in rule with a pseudo-observation: $\hat{\sigma}^2_0 = 1/4$ and, for $i \geq 1$, $\hat{\mu}_i = (1/2 + \sum_{j \leq i} Y_j)/(i + 1)$ and $\hat{\sigma}^2_i = (1/4 + \sum_{j \leq i}(Y_j - \hat{\mu}_j)^2)/(i + 1)$. The bets use only values already seen and do not depend on m; validity needs only that, but power, and hence the feasibility numbers below, depends on this choice. Its final value K_n has expectation at most 1: it is an e-value. The order must be random: a random set visited in a fixed order does not have these conditional means. Rather than rely on one random order, we draw $R = 100$ independent orders and average their final values, $\bar{K}(m) = R^{-1}\sum_r K_n^{(r)}$. An average of e-values is an e-value [59], so $p(m) = \min(1, 1/\bar{K}(m))$ satisfies $P(p(m) \leq \delta') \leq \delta'$ under the null by Markov's inequality. Each $K_n^{(r)}$ increases with m, so $\bar{K}$ does too, and rejecting at m^* implies rejecting at every larger m. If the remainder's error exceeds ε, then $\mu > m^*$, so certifying t requires rejecting at $m = \mu$, a true null, which happens with probability at most δ'. Testing each of the 16 thresholds at $\delta' = \delta/16$ gives the proposition in Section 3.4. The only randomness used is the choice of clusters to hand-code and the visiting orders; dependence among errors is unrestricted. The orders come from a pseudo-random generator with a recorded seed, fixed before this analysis and not chosen by any outcome.

LTT on i.i.d. clusters. For the population target, the same Y_i are tested against the constant null mean ε with the same betting process and the same averaging over orders. This is valid if clusters are independent draws whose weights never exceed B; exchangeability alone is not enough, because a shared latent cause can make every cluster right or every cluster wrong together. For MFRC, B is a genuine bound (eight codes per comment at one weight); for transcripts of unbounded length it is not.

Learned scores and pipelines. Per-code maps and the prevalence baseline are fitted on half of the calibration clusters and frozen; corpus LTT runs on the other half, with the corpus taken as that half plus the test split, and the frozen scores are applied to the test split. A two-stage cascade is one hypothesis per combination of stage thresholds (8 per stage, plus "off" for the second), with Bonferroni over all 72 combinations; the certified combination with the largest calibration coverage is used.

What the procedure needs. For a coder that accepts everything and is never wrong, $Y_i = \varepsilon(1 - r_i)$ with r_i the unit's weight relative to B, and the conditional null mean at m^* is at most ε. Since bets are capped at 1, each unit can multiply the capital by at most $1 + \varepsilon r_i$, whatever the order, and certification with the grid needs the capital to reach $16/0.05 = 320$. Hence $\sum_{\text{coded}} \log(1 + \varepsilon r_i) \geq \log 320$ is necessary for any set of hand-coded units and any visiting order (Table 21, "lower bound"). For MFRC, whose comments all have the same weight, at least 119 comments are needed; our procedure first certifies after a median of 123. With whole transcripts as units, where a typical transcript carries 8–27% of the largest one's weight, the bound rules out certification on AnnoMI (both tasks) and TalkMoves student even if every transcript but one were

hand-coded, and on TalkMoves teacher it needs at least 388 of 444 lessons. These requirements belong to this procedure (capped bets, scaling by the largest unit, Bonferroni over the grid) under whole-transcript sampling. Because corpus LTT needs only random selection, not independence, the unit can be smaller: drawing single utterances or sentences at random, with their context, keeps every unit's weight close to B, and a perfect coder is then certified after a median of 139–163 hand-coded segments (Table 21). Table 22 applies that design to the real coders.

C Additional results

C.1 RQ2: uncertainty and human disagreement, in detail

These tables give the per-coder numbers behind Figure 4 (§4.2): how strongly each coder's uncertainty tracks the spread of human labels, and how its agreement with annotators compares with each annotator's agreement with the others. The decision models and the verbalized signals track disagreement; token probabilities and vote shares do not. No coder falls below the human range, which is broad, so the comparison is descriptive.

Table 10: RQ2: model uncertainty and human disagreement, by coder (test items with ≥ 3 annotators)

task	signal	Spearman: uncertainty vs human split	AUROC: flags split items	model κ (mean over references)	annotators below model
MFRC	Jev	0.326	0.718	0.269	2 of 6
MFRC	OpenAI Decisions	0.242	0.659	0.260	2 of 6
MFRC	Luna verbal	0.339	0.721	0.250	2 of 6
MFRC	Haiku verbal	0.375	0.742	0.189	1 of 6
MFRC	gpt-oss verbal	0.307	0.673	0.310	3 of 6
MFRC	Luna logprob	0.138	0.582	0.250	2 of 6
MFRC	3-LLM agreement	0.199	0.591	0.280	2 of 6
MFRC	Sonnet verbal	0.427	0.784	0.361	5 of 6
MFRC	Clef (27B)	0.404	0.771	0.282	2 of 6
MFRC	Clef-flash (9B)	0.395	0.764	0.306	3 of 6
MFRC	pplx-decider	0.398	0.767	0.246	1 of 6
MFRC	Kev-9B	0.325	0.717	0.239	2 of 6
MFRC	Kev-27B	0.394	0.764	0.249	2 of 6
AnnoMI client	Jev	0.547	0.859	0.706	7 of 10
AnnoMI client	OpenAI Decisions	0.517	0.812	0.696	7 of 10
AnnoMI client	Luna verbal	0.514	0.803	0.701	7 of 10
AnnoMI client	Haiku verbal	0.568	0.875	0.590	3 of 10
AnnoMI client	gpt-oss verbal	0.522	0.817	0.655	7 of 10
AnnoMI client	Luna logprob	-0.100	0.462	0.701	7 of 10
AnnoMI client	3-LLM agreement	0.324	0.604	0.648	7 of 10
AnnoMI client	Sonnet verbal	0.647	0.872	0.714	9 of 10
AnnoMI client	Clef (27B)	0.636	0.904	0.508	1 of 10
AnnoMI client	Clef-flash (9B)	0.600	0.912	0.483	1 of 10
AnnoMI client	pplx-decider	0.652	0.864	0.738	8 of 10
AnnoMI client	Kev-9B	0.179	0.627	0.462	1 of 10
AnnoMI client	Kev-27B	0.572	0.872	0.672	7 of 10
AnnoMI therapist	Jev	0.574	0.773	0.891	10 of 10
AnnoMI therapist	OpenAI Decisions	0.438	0.729	0.914	10 of 10
AnnoMI therapist	Luna verbal	0.560	0.797	0.899	10 of 10
AnnoMI therapist	Haiku verbal	0.359	0.670	0.834	6 of 10
AnnoMI therapist	gpt-oss verbal	0.379	0.648	0.880	8 of 10
AnnoMI therapist	Luna logprob	-0.030	0.490	0.899	10 of 10
AnnoMI therapist	3-LLM agreement	0.208	0.570	0.898	10 of 10
AnnoMI therapist	Sonnet verbal	0.511	0.738	0.883	8 of 10
AnnoMI therapist	Clef (27B)	0.327	0.664	0.807	4 of 10
AnnoMI therapist	Clef-flash (9B)	0.346	0.691	0.809	4 of 10
AnnoMI therapist	pplx-decider	0.542	0.771	0.908	10 of 10
AnnoMI therapist	Kev-9B	0.435	0.715	0.721	1 of 10
AnnoMI therapist	Kev-27B	0.505	0.766	0.793	3 of 10

Model κ: against the majority of the other annotators, averaged over the leave-one-annotator-out references. The human range for each task is in Table 11.

Table 11: RQ2: agreement of each annotator with the majority of the other annotators (the human range)

task	items	annotators	human κ min	human κ median	human κ max
MFRC	1382	6	0.110	0.295	0.372
AnnoMI client	91	10	0.414	0.607	0.773
AnnoMI therapist	93	10	0.631	0.834	0.886

Leave-one-annotator-out κ on the same items as Table 10. AnnoMI's multi-annotated items come from four transcripts.

C.2 RQ3: robustness by arm and task

Each perturbation was run on identical test items and compared with the canonical request (§4.3). The first table gives every arm's flip rate and its effect on agreement, calibration and ranking; the second gives per-task intervals for the two pre-specified outcomes on the arms that matter most, reversed options and removed context. Option order moves Kev-9B far more than any other coder, and removing the previous turn lowers agreement for every coder.

Table 12: RQ3: robustness to perturbations (design-weighted within tasks; equal-weight mean over the tasks each arm covers; vs the canonical arm on identical items)

signal	arm	tasks	flip rate	ΔAURC	$\Delta\kappa$	Δ error AUROC	ΔECE	flips at low confidence (AUROC)
Clef-flash (9B)	one code per call	1	0.040	-0.001	-0.074	-0.003	-0.055	0.935
Clef-flash (9B)	neutral labels	5	0.094	-0.008	0.006	0.004	-0.004	0.881
Clef-flash (9B)	no previous turn	4	0.236	0.026	-0.016	-0.026	0.000	0.768
Clef-flash (9B)	paraphrase A	5	0.060	0.001	-0.005	-0.001	0.005	0.908
Clef-flash (9B)	paraphrase B	5	0.096	0.010	-0.019	-0.001	-0.010	0.877
Clef-flash (9B)	options reversed	5	0.008	-0.000	-0.002	0.003	-0.002	0.918
Clef-flash (9B)	shuffle 1	5	0.004	-0.000	0.000	0.002	-0.002	0.953
Clef-flash (9B)	shuffle 2	5	0.004	-0.000	0.001	0.001	-0.002	0.963
Clef-flash (9B)	shuffle 3	5	0.004	-0.000	-0.001	0.002	-0.002	0.954
Jev	one code per call	1	0.008	-0.000	-0.004	0.000	-0.001	0.988
Jev	neutral labels	5	0.050	-0.003	-0.003	0.007	0.002	0.943
Jev	no previous turn	4	0.151	0.037	-0.038	-0.025	0.021	0.816
Jev	paraphrase A	5	0.032	-0.000	-0.006	0.011	-0.001	0.952
Jev	paraphrase B	5	0.036	0.006	-0.007	-0.003	0.001	0.946
Jev	options reversed	5	0.042	0.006	-0.005	0.000	-0.005	0.951
Jev	shuffle 1	5	0.042	0.001	-0.002	0.003	-0.003	0.949
Jev	shuffle 2	5	0.040	0.000	0.000	0.001	-0.005	0.952
Jev	shuffle 3	5	0.041	0.001	0.000	0.001	-0.000	0.950
Kev-9B	one code per call	1	0.002	-0.000	-0.003	0.000	-0.000	0.997
Kev-9B	neutral labels	3	0.102	0.000	-0.000	0.007	-0.013	0.899
Kev-9B	no previous turn	2	0.175	0.010	-0.012	-0.016	-0.020	0.747
Kev-9B	paraphrase A	3	0.064	0.008	-0.021	0.012	0.006	0.926
Kev-9B	paraphrase B	3	0.076	0.004	-0.020	0.009	0.012	0.920
Kev-9B	options reversed	4	0.306	0.119	-0.099	0.046	0.044	0.813
Kev-9B	shuffle 1	4	0.192	0.066	-0.042	-0.006	-0.026	0.842
Kev-9B	shuffle 2	4	0.204	0.072	-0.053	0.007	-0.034	0.843
Kev-9B	shuffle 3	3	0.129	0.035	-0.023	-0.014	-0.002	0.864
Luna verbal	neutral labels	5	0.081	0.016	-0.006	-0.038	0.006	0.805
Luna verbal	options reversed	5	0.100	-0.011	0.031	-0.022	-0.015	0.783
Luna verbal	shuffle 1	5	0.106	-0.004	0.025	-0.026	-0.012	0.785
OpenAI Decisions	one code per call	1	0.000	0.000	0.000	0.000	0.000	–
OpenAI Decisions	neutral labels	5	0.096	-0.005	-0.004	0.001	0.008	0.889
OpenAI Decisions	no previous turn	4	0.188	0.011	-0.040	-0.002	-0.016	0.794
OpenAI Decisions	paraphrase A	5	0.072	-0.002	0.012	-0.005	0.035	0.899
OpenAI Decisions	paraphrase B	5	0.076	0.001	0.017	-0.024	0.026	0.895
OpenAI Decisions	options reversed	5	0.103	-0.005	0.021	-0.014	0.022	0.873
OpenAI Decisions	shuffle 1	5	0.089	-0.005	0.012	-0.001	0.018	0.875
OpenAI Decisions	shuffle 2	5	0.083	-0.004	0.021	-0.018	0.015	0.861
OpenAI Decisions	shuffle 3	5	0.088	0.000	0.017	-0.019	0.015	0.870
pplx-decider	one code per call	1	0.000	0.000	0.000	0.000	0.000	–
pplx-decider	neutral labels	5	0.071	-0.004	-0.002	0.009	0.012	0.890
pplx-decider	no previous turn	4	0.154	0.023	-0.041	-0.012	0.029	0.801
pplx-decider	paraphrase A	5	0.038	-0.001	0.008	-0.007	-0.002	0.932
pplx-decider	paraphrase B	5	0.037	-0.001	0.006	-0.003	0.000	0.930
pplx-decider	options reversed	5	0.064	-0.019	0.012	0.021	0.010	0.875
pplx-decider	shuffle 1	5	0.064	-0.012	-0.002	0.014	0.009	0.853
pplx-decider	shuffle 2	5	0.062	-0.013	0.006	0.009	0.007	0.852
pplx-decider	shuffle 3	5	0.065	-0.012	0.003	0.013	0.008	0.850
Sonnet verbal	neutral labels	5	0.049	-0.002	-0.016	0.011	0.004	0.915
Sonnet verbal	options reversed	5	0.052	-0.003	-0.033	0.016	0.012	0.882
Sonnet verbal	shuffle 1	5	0.059	0.000	-0.020	0.006	-0.002	0.887

Flip rate and ΔAURC are the pre-specified primary outcomes; Table 13 gives per-task 95% intervals for selected arms. 'No previous turn' removes information rather than rewording the request.

Table 13: RQ3 primary outcomes with per-task 95% intervals (cluster bootstrap) for two arms

arm	signal	task	flip rate [95% CI]	ΔAURC [95% CI]
no previous turn	Clef-flash (9B)	AnnoMI client	0.130 [0.104, 0.161]	+0.031 [+0.011, +0.055]
no previous turn	Clef-flash (9B)	AnnoMI therapist	0.121 [0.090, 0.151]	+0.008 [-0.002, +0.020]
no previous turn	Clef-flash (9B)	TalkMoves student	0.306 [0.279, 0.336]	+0.117 [+0.085, +0.151]
no previous turn	Clef-flash (9B)	TalkMoves teacher	0.385 [0.356, 0.416]	-0.052 [-0.085, -0.013]
no previous turn	Jev	AnnoMI client	0.082 [0.065, 0.102]	+0.033 [+0.015, +0.051]
no previous turn	Jev	AnnoMI therapist	0.088 [0.068, 0.105]	-0.001 [-0.013, +0.013]
no previous turn	Jev	TalkMoves student	0.227 [0.204, 0.253]	+0.081 [+0.059, +0.105]
no previous turn	Jev	TalkMoves teacher	0.208 [0.184, 0.231]	+0.033 [+0.013, +0.051]
no previous turn	Kev-9B	AnnoMI client	0.216 [0.189, 0.246]	+0.005 [-0.027, +0.035]
no previous turn	Kev-9B	AnnoMI therapist	0.135 [0.112, 0.159]	+0.014 [-0.002, +0.031]
no previous turn	OpenAI Decisions	AnnoMI client	0.076 [0.059, 0.096]	+0.023 [+0.003, +0.044]
no previous turn	OpenAI Decisions	AnnoMI therapist	0.106 [0.085, 0.124]	+0.012 [+0.001, +0.024]
no previous turn	OpenAI Decisions	TalkMoves student	0.342 [0.315, 0.371]	+0.011 [-0.023, +0.045]
no previous turn	OpenAI Decisions	TalkMoves teacher	0.230 [0.203, 0.257]	-0.004 [-0.026, +0.019]
no previous turn	pplx-decider	AnnoMI client	0.086 [0.068, 0.110]	+0.032 [+0.010, +0.052]
no previous turn	pplx-decider	AnnoMI therapist	0.058 [0.041, 0.074]	+0.004 [-0.006, +0.014]
no previous turn	pplx-decider	TalkMoves student	0.311 [0.282, 0.345]	+0.061 [+0.037, +0.084]
no previous turn	pplx-decider	TalkMoves teacher	0.163 [0.141, 0.186]	-0.006 [-0.024, +0.013]
options reversed	Clef-flash (9B)	AnnoMI client	0.000 [0.000, 0.000]	+0.000 [+0.000, +0.000]
options reversed	Clef-flash (9B)	AnnoMI therapist	0.000 [0.000, 0.000]	+0.000 [+0.000, +0.000]
options reversed	Clef-flash (9B)	MFRC	0.040 [0.037, 0.043]	-0.001 [-0.002, -0.001]
options reversed	Clef-flash (9B)	TalkMoves student	0.000 [0.000, 0.000]	+0.000 [+0.000, +0.000]
options reversed	Clef-flash (9B)	TalkMoves teacher	0.000 [0.000, 0.000]	+0.000 [+0.000, +0.000]
options reversed	Jev	AnnoMI client	0.035 [0.023, 0.044]	+0.002 [-0.003, +0.008]
options reversed	Jev	AnnoMI therapist	0.021 [0.013, 0.029]	-0.008 [-0.015, -0.001]
options reversed	Jev	MFRC	0.009 [0.007, 0.011]	+0.000 [-0.000, +0.000]
options reversed	Jev	TalkMoves student	0.072 [0.057, 0.087]	+0.034 [+0.024, +0.044]
options reversed	Jev	TalkMoves teacher	0.072 [0.056, 0.088]	+0.001 [-0.007, +0.008]
options reversed	Kev-9B	AnnoMI client	0.444 [0.400, 0.483]	+0.180 [+0.137, +0.232]
options reversed	Kev-9B	AnnoMI therapist	0.224 [0.193, 0.255]	+0.067 [+0.038, +0.097]
options reversed	Kev-9B	MFRC	0.002 [0.001, 0.003]	-0.000 [-0.000, +0.000]
options reversed	Kev-9B	TalkMoves teacher	0.555 [0.522, 0.588]	+0.230 [+0.193, +0.268]
options reversed	OpenAI Decisions	AnnoMI client	0.064 [0.050, 0.079]	+0.014 [-0.011, +0.035]
options reversed	OpenAI Decisions	AnnoMI therapist	0.086 [0.067, 0.105]	+0.020 [+0.004, +0.035]
options reversed	OpenAI Decisions	MFRC	0.000 [0.000, 0.000]	+0.000 [+0.000, +0.000]
options reversed	OpenAI Decisions	TalkMoves student	0.212 [0.188, 0.236]	-0.047 [-0.070, -0.023]
options reversed	OpenAI Decisions	TalkMoves teacher	0.152 [0.129, 0.174]	-0.013 [-0.030, +0.004]
options reversed	Sonnet verbal	AnnoMI client	0.040 [0.005, 0.074]	-0.003 [-0.021, +0.019]
options reversed	Sonnet verbal	AnnoMI therapist	0.051 [0.012, 0.105]	+0.001 [-0.006, +0.014]
options reversed	Sonnet verbal	MFRC	0.050 [0.032, 0.069]	+0.003 [+0.001, +0.007]
options reversed	Sonnet verbal	TalkMoves student	0.038 [0.000, 0.084]	+0.001 [-0.018, +0.023]
options reversed	Sonnet verbal	TalkMoves teacher	0.082 [0.026, 0.145]	-0.017 [-0.050, +0.008]
options reversed	pplx-decider	AnnoMI client	0.027 [0.016, 0.042]	-0.006 [-0.014, +0.001]
options reversed	pplx-decider	AnnoMI therapist	0.039 [0.022, 0.057]	-0.001 [-0.005, +0.002]
options reversed	pplx-decider	MFRC	0.000 [0.000, 0.000]	+0.000 [+0.000, +0.000]
options reversed	pplx-decider	TalkMoves student	0.122 [0.104, 0.142]	-0.016 [-0.035, +0.003]
options reversed	pplx-decider	TalkMoves teacher	0.129 [0.113, 0.148]	-0.073 [-0.094, -0.051]

Positive ΔAURC means a larger area under the risk–coverage curve for the perturbed run (worse selective risk); AURC reflects the error rate as well as the ranking of errors.

C.3 EQ1: calibration and recalibration

To separate calibration from ranking (§5.1), we recalibrated every signal per task on the calibration split and re-scored it on the test split. Platt and isotonic maps bring calibration error down to 0.03–0.07, and the fitted Platt map, which is increasing for every decision model and verbalized signal, leaves their ranking unchanged, while temperature scaling depends on how exact zeros and ones are floored, so we draw no conclusion from it. The fixed rule "accept ≥0.95" shows what raw probabilities do when taken at face value.

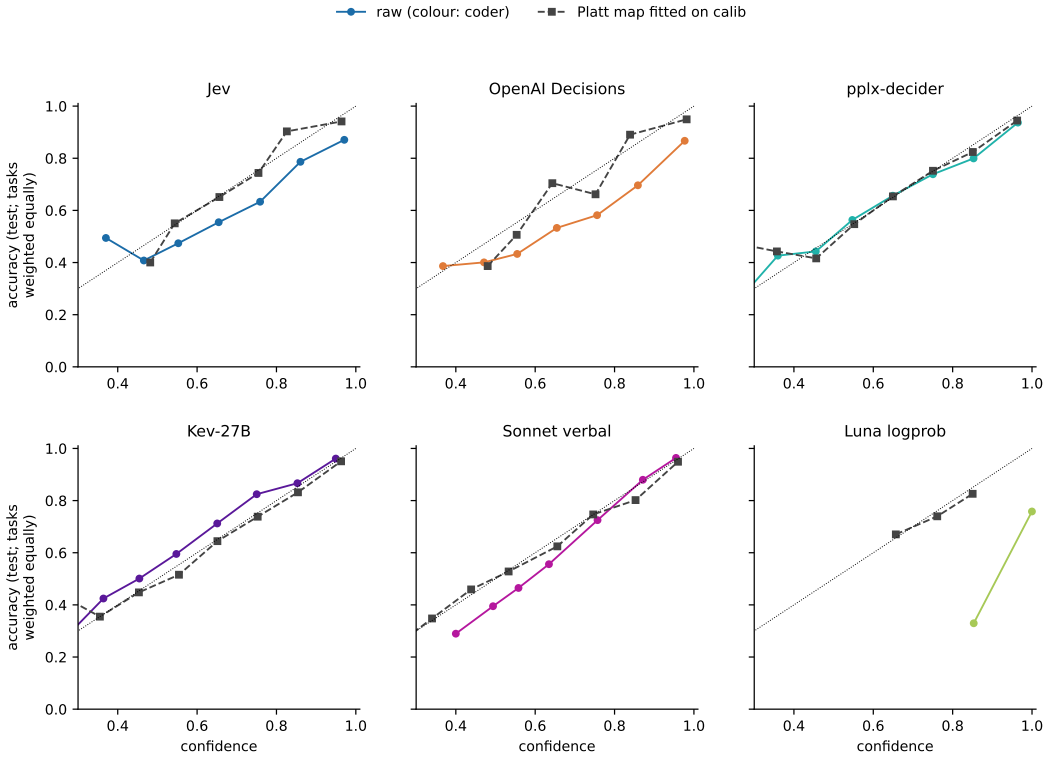


Figure 9: EQ1: reliability on the test split before (coloured) and after (dashed grey) a Platt map fitted per task on the calibration split, for four decision models, Sonnet's verbalized confidence and Luna's token probabilities. Tasks are weighted equally; bins with fewer than 20 decisions are omitted.

Table 14: EQ1 (exploratory): recalibration and the fixed rule 'accept if confidence ≥ 0.95' (pooled)

A. Expected calibration error, raw and after recalibration

signal	ECE raw	ECE Platt	ECE isotonic	ECE temperature
Jev	0.120	0.038	0.040	0.120
OpenAI Decisions	0.122	0.073	0.042	0.098
Clef (27B)	0.055	0.041	0.045	0.054
Clef-flash (9B)	0.065	0.043	0.035	0.051
pplx-decider	0.052	0.043	0.048	0.047
Kev-9B	0.100	0.041	0.053	0.069
Kev-27B	0.073	0.032	0.037	0.029
Luna verbal	0.180	0.047	0.040	0.059
Haiku verbal	0.154	0.045	0.035	0.075
gpt-oss verbal	0.204	0.056	0.038	0.075
Qwen3.5-9B verbal	0.303	0.045	0.037	0.099
Luna logprob	0.243	0.038	0.039	0.106
gpt-oss logprob	0.284	0.040	0.045	0.033
Qwen3.5-9B logprob	0.334	0.052	0.054	0.060
Sonnet verbal	0.051	0.034	0.039	0.053
Qwen3.8-2.4T verbal	0.102	0.060	0.061	0.091

B. Error ranking, and the fixed rule 'accept if confidence ≥ 0.95' applied to raw confidence

signal	error AUROC raw	error AUROC Platt	share accepted by '≥ 0.95'	its error
Jev	0.756	0.756	0.364	0.121
OpenAI Decisions	0.736	0.736	0.498	0.115
Clef (27B)	0.721	0.721	0.081	0.007
Clef-flash (9B)	0.712	0.712	0.043	0.001
pplx-decider	0.764	0.764	0.281	0.041
Kev-9B	0.694	0.694	0.041	0.032
Kev-27B	0.788	0.788	0.155	0.034
Luna verbal	0.741	0.741	0.635	0.139
Haiku verbal	0.783	0.783	0.310	0.044
gpt-oss verbal	0.727	0.727	0.523	0.120
Qwen3.5-9B verbal	0.646	0.646	0.966	0.331
Luna logprob	0.558	0.556	0.994	0.241
gpt-oss logprob	0.558	0.539	0.998	0.285
Qwen3.5-9B logprob	0.684	0.684	0.984	0.334
Sonnet verbal	0.801	0.801	0.225	0.021
Qwen3.8-2.4T verbal	0.802	0.802	0.259	0.020

Maps fitted per task on calib. Platt and isotonic maps act on the decision's confidence; temperature scaling acts on the full distribution and is sensitive to how exact 0s and 1s are floored (Table 15; here 1e-12). Errors are among accepted test decisions.

Table 15: Temperature scaling and per-code maps depend on how exact 0s and 1s are floored before taking logarithms (pooled)

A. ECE

signal	raw	temperature, floor 1e-4	temperature, floor 1e-12
Jev	0.120	0.046	0.120
Kev-27B	0.073	0.029	0.029
Luna logprob	0.243	0.040	0.106
OpenAI Decisions	0.122	0.040	0.098
gpt-oss logprob	0.284	0.033	0.033
Qwen3.5-9B logprob	0.334	0.055	0.060
Sonnet verbal	0.051	0.053	0.053

B. Error AUROC

signal	raw	temperature, floor 1e-4	temperature, floor 1e-12	per-code Platt, floor 1e-4	per-code Platt, floor 1e-12
Jev	0.756	0.757	0.749	0.809	0.797
Kev-27B	0.788	0.788	0.788	0.814	0.814
Luna logprob	0.558	0.489	0.579	0.670	0.671
OpenAI Decisions	0.736	0.735	0.735	0.785	0.771
gpt-oss logprob	0.558	0.495	0.542	0.698	0.717
Qwen3.5-9B logprob	0.684	0.665	0.666	0.733	0.757
Sonnet verbal	0.801	0.802	0.802	0.826	0.826

One temperature per task, fitted by weighted log-loss on calib. A task-level Platt or isotonic map is monotone and leaves error ranking unchanged at any floor; per-code Platt maps (EQ3) are not monotone across codes, so the floor changes their ranking for coders that return exact 0s and 1s.

C.4 EQ2: threshold rules and the cost of a guarantee

These analyses test whether each threshold rule keeps its 5% target and what a valid guarantee costs (§5.2). The re-split tables repeat the random calibration/test split 500 times per task: the naive rule exceeds 5% in about half of them on MFRC and LTT on decisions in more than the nominal 5%, while corpus LTT never does. The budget curves and the single-threshold variant show how much hand-coding a guarantee needs on MFRC. The simulation checks each rule against a known true risk under correlated errors, and the feasibility and utterance-sampling tables show why dialogue certifies nothing under whole-transcript sampling, and why sampling single utterances removes that obstacle yet still yields almost no certificates for the real coders.

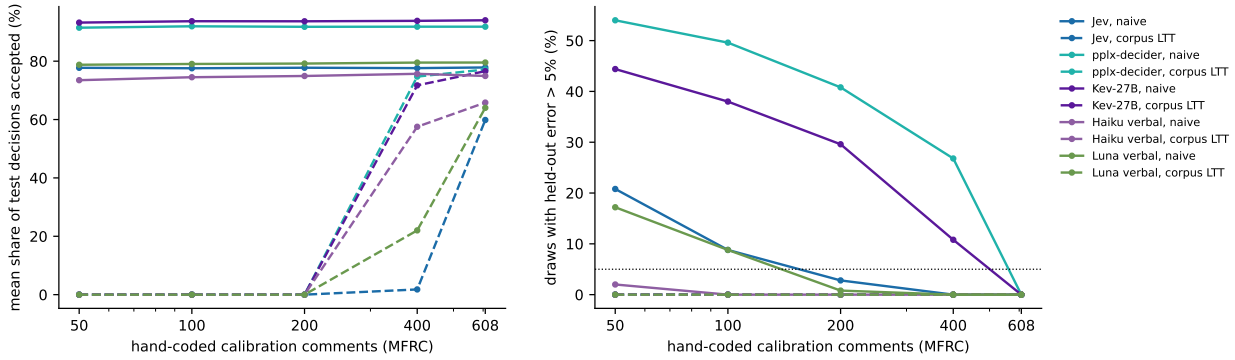


Figure 10: EQ2: calibration budget on MFRC. Left: mean share of test decisions accepted; right: share of draws whose held-out error exceeds 5%, when only n randomly drawn comments are hand-coded (250 draws; 608 is the full calibration split). Solid: naive rule; dashed: corpus LTT. Corpus LTT certifies nothing for any plotted coder with 200 or fewer comments, and 2% (Jev) to 75% (pplx-decider) with 400. For each draw the corpus is the drawn comments plus the test split, whose decisions are the ones scored. The dialogue tasks are omitted: with whole transcripts hand-coded, corpus LTT certifies nothing there.

Table 16: EQ2 (exploratory): threshold rules at a 5% target (percent)

A. Frozen split, pooled over tasks

signal	naive: accepted	naive: error	corpus LTT: accepted	corpus LTT: error	oracle: accepted
Jev	32.9	6.8	12.0	1.6	16.3
OpenAI Decisions	22.7	4.7	0.0	–	17.8
Clef (27B)	24.0	5.8	16.5	3.0	20.3
Clef-flash (9B)	25.4	7.0	15.6	2.7	20.7
pplx-decider	40.8	7.5	15.4	2.3	32.1
Kev-9B	23.9	6.3	13.4	2.4	19.2
Kev-27B	44.2	7.0	15.3	2.1	37.2
Luna verbal	20.2	4.6	12.8	2.2	16.2
Haiku verbal	16.1	3.2	13.2	1.7	16.8
gpt-oss verbal	22.1	5.6	0.0	–	20.6
Qwen3.5-9B verbal	3.4	11.4	0.0	–	0.0
Luna logprob	0.0	–	0.0	–	11.8
gpt-oss logprob	1.2	29.7	0.0	–	0.0
Qwen3.5-9B logprob	9.3	7.9	0.0	–	8.2
Sonnet verbal	43.4	6.2	16.5	2.5	40.1
Qwen3.8-2.4T verbal	37.9	4.3	0.0	–	38.0

B. MFRC: share of re-splits with held-out error above 5%, and corpus LTT with per-code maps

signal	naive	LTT on decisions	LTT on i.i.d. clusters	corpus LTT	per-code maps, corpus LTT: accepted
Jev	49.4	6.8	0.0	0.0	0.0
OpenAI Decisions	28.0	5.8	0.0	0.0	0.0
Clef (27B)	47.0	8.4	0.0	0.0	75.4
Clef-flash (9B)	51.8	8.4	0.0	0.0	65.5
pplx-decider	50.0	11.2	0.0	0.0	72.0
Kev-9B	52.4	9.0	0.0	0.0	0.0
Kev-27B	47.0	9.2	0.0	0.0	0.0
Luna verbal	5.2	0.0	0.0	0.0	0.0
Haiku verbal	19.4	6.0	0.0	0.0	0.0
gpt-oss verbal	0.0	0.0	0.0	0.0	0.0
Qwen3.5-9B verbal	0.0	0.0	0.0	0.0	0.0
Luna logprob	0.0	0.0	0.0	0.0	0.0
gpt-oss logprob	15.2	0.0	0.0	0.0	0.0
Qwen3.5-9B logprob	47.2	5.2	0.0	0.0	0.0
Sonnet verbal	6.0	0.0	0.0	0.0	68.9
Qwen3.8-2.4T verbal	5.2	0.6	0.0	0.0	0.0

Panel A: frozen split, equal-weight means over the five tasks; oracle: fitted on test. Panel B: 500 random cluster re-splits of MFRC; for corpus LTT each test split is the uncoded remainder of its re-split, so exceedance there is a direct check of its guarantee. Per-code maps are fitted on half of the calib clusters, frozen, certified on the other half and applied unchanged to test.

Table 17: Naive rule and LTT on decisions by task over 500 cluster re-splits (percent)

signal	task	naive: accepted	naive: none accepted	naive: error > 5%	LTT on decisions: accepted	LTT on decisions: none accepted	LTT on decisions: error > 5%
Clef (27B)	AnnoMI client	7.3	5.0	73.8	0.0	100.0	0.0
Clef (27B)	AnnoMI therapist	18.8	0.0	63.6	0.2	99.4	0.6
Clef (27B)	MFRC	90.2	0.0	47.0	88.3	0.0	8.4
Clef (27B)	TalkMoves student	1.4	0.2	65.4	0.0	100.0	0.0
Clef (27B)	TalkMoves teacher	0.6	0.8	67.8	0.0	100.0	0.0
Clef-flash (9B)	AnnoMI client	11.5	1.2	84.8	0.0	100.0	0.0
Clef-flash (9B)	AnnoMI therapist	10.8	8.2	56.4	0.0	100.0	0.0
Clef-flash (9B)	MFRC	89.8	0.0	51.8	87.1	0.0	8.4
Clef-flash (9B)	TalkMoves student	2.4	0.0	33.2	0.0	100.0	0.0
Clef-flash (9B)	TalkMoves teacher	0.4	6.6	53.2	0.0	100.0	0.0
Jev	AnnoMI client	3.1	91.0	9.0	0.0	100.0	0.0
Jev	AnnoMI therapist	18.0	62.8	37.2	0.1	99.8	0.2
Jev	MFRC	80.8	0.0	49.4	78.3	0.0	6.8
Jev	TalkMoves student	1.7	85.0	15.0	0.0	100.0	0.0
Jev	TalkMoves teacher	2.5	81.0	19.0	0.0	100.0	0.0
Kev-27B	AnnoMI client	15.6	0.2	70.2	0.0	100.0	0.0
Kev-27B	AnnoMI therapist	51.9	0.0	41.8	12.4	68.0	2.0
Kev-27B	MFRC	94.0	0.0	47.0	91.7	0.0	9.2
Kev-27B	TalkMoves student	20.3	0.0	62.8	0.0	100.0	0.0
Kev-27B	TalkMoves teacher	21.9	0.6	72.2	0.0	100.0	0.0
Kev-9B	AnnoMI client	2.8	0.4	68.8	0.0	100.0	0.0
Kev-9B	AnnoMI therapist	14.2	0.0	79.8	0.1	99.6	0.4
Kev-9B	MFRC	87.8	0.0	52.4	56.8	16.0	9.0
Kev-9B	TalkMoves student	0.2	67.0	33.0	0.0	100.0	0.0
Kev-9B	TalkMoves teacher	0.4	5.6	41.6	0.0	100.0	0.0
OpenAI Decisions	AnnoMI client	1.8	91.8	8.2	0.0	100.0	0.0
OpenAI Decisions	AnnoMI therapist	32.1	18.8	51.2	1.4	96.8	3.2
OpenAI Decisions	MFRC	74.3	0.0	28.0	69.6	0.0	5.8
OpenAI Decisions	TalkMoves student	0.9	30.6	69.4	0.0	100.0	0.0
OpenAI Decisions	TalkMoves teacher	0.1	99.6	0.4	0.0	100.0	0.0
pplx-decider	AnnoMI client	17.3	0.0	65.0	0.0	100.0	0.0
pplx-decider	AnnoMI therapist	58.8	0.0	53.4	9.1	80.0	2.6
pplx-decider	MFRC	91.5	0.0	50.0	89.2	0.0	11.2
pplx-decider	TalkMoves student	13.1	0.0	52.0	0.0	100.0	0.0
pplx-decider	TalkMoves teacher	3.4	44.0	56.0	0.0	100.0	0.0
Sonnet verbal	AnnoMI client	12.7	0.2	49.2	0.2	99.2	0.8
Sonnet verbal	AnnoMI therapist	56.0	0.0	38.6	7.7	83.8	3.0
Sonnet verbal	MFRC	89.8	0.0	6.0	88.9	0.0	0.0
Sonnet verbal	TalkMoves student	11.6	10.6	62.0	0.0	100.0	0.0
Sonnet verbal	TalkMoves teacher	39.0	0.0	39.2	1.8	95.4	4.2

Accepted: mean share of test decisions accepted; none accepted: share of re-splits accepting nothing; error > 5%: share of re-splits whose held-out error exceeds 5%. A re-split that accepts nothing cannot exceed 5%; read exceedance with acceptance.

Table 18: LTT on i.i.d. clusters and corpus LTT by task over 500 cluster re-splits (percent)

signal	task	i.i.d.-cluster LTT: accepted	i.i.d.-cluster LTT: none accepted	i.i.d.-cluster LTT: error > 5%	corpus LTT: accepted	corpus LTT: none accepted	corpus LTT: error > 5%
Clef (27B)	AnnoMI client	0.0	100.0	0.0	0.0	100.0	0.0
Clef (27B)	AnnoMI therapist	0.0	100.0	0.0	0.0	100.0	0.0
Clef (27B)	MFRC	82.3	0.0	0.0	81.0	0.0	0.0
Clef (27B)	TalkMoves student	0.0	100.0	0.0	0.0	100.0	0.0
Clef (27B)	TalkMoves teacher	0.0	100.0	0.0	0.0	100.0	0.0
Clef-flash (9B)	AnnoMI client	0.0	100.0	0.0	0.0	100.0	0.0
Clef-flash (9B)	AnnoMI therapist	0.0	100.0	0.0	0.0	100.0	0.0
Clef-flash (9B)	MFRC	79.2	0.0	0.0	77.5	0.0	0.0
Clef-flash (9B)	TalkMoves student	0.0	100.0	0.0	0.0	100.0	0.0
Clef-flash (9B)	TalkMoves teacher	0.0	100.0	0.0	0.0	100.0	0.0
Jev	AnnoMI client	0.0	100.0	0.0	0.0	100.0	0.0
Jev	AnnoMI therapist	0.0	100.0	0.0	0.0	100.0	0.0
Jev	MFRC	69.7	0.0	0.0	67.5	0.0	0.0
Jev	TalkMoves student	0.0	100.0	0.0	0.0	100.0	0.0
Jev	TalkMoves teacher	0.0	100.0	0.0	0.0	100.0	0.0
Kev-27B	AnnoMI client	0.0	100.0	0.0	0.0	100.0	0.0
Kev-27B	AnnoMI therapist	0.0	100.0	0.0	0.0	100.0	0.0
Kev-27B	MFRC	84.8	0.0	0.0	82.4	0.0	0.0
Kev-27B	TalkMoves student	0.0	100.0	0.0	0.0	100.0	0.0
Kev-27B	TalkMoves teacher	0.0	100.0	0.0	0.0	100.0	0.0
Kev-9B	AnnoMI client	0.0	100.0	0.0	0.0	100.0	0.0
Kev-9B	AnnoMI therapist	0.0	100.0	0.0	0.0	100.0	0.0
Kev-9B	MFRC	75.9	0.0	0.0	72.3	0.6	0.0
Kev-9B	TalkMoves student	0.0	100.0	0.0	0.0	100.0	0.0
Kev-9B	TalkMoves teacher	0.0	100.0	0.0	0.0	100.0	0.0
OpenAI Decisions	AnnoMI client	0.0	100.0	0.0	0.0	100.0	0.0
OpenAI Decisions	AnnoMI therapist	0.0	100.0	0.0	0.0	100.0	0.0
OpenAI Decisions	MFRC	14.8	67.0	0.0	0.4	99.0	0.0
OpenAI Decisions	TalkMoves student	0.0	100.0	0.0	0.0	100.0	0.0
OpenAI Decisions	TalkMoves teacher	0.0	100.0	0.0	0.0	100.0	0.0
pplx-decider	AnnoMI client	0.0	100.0	0.0	0.0	100.0	0.0
pplx-decider	AnnoMI therapist	0.0	100.0	0.0	0.0	100.0	0.0
pplx-decider	MFRC	82.7	0.0	0.0	81.1	0.0	0.0
pplx-decider	TalkMoves student	0.0	100.0	0.0	0.0	100.0	0.0
pplx-decider	TalkMoves teacher	0.0	100.0	0.0	0.0	100.0	0.0
Sonnet verbal	AnnoMI client	0.0	100.0	0.0	0.0	100.0	0.0
Sonnet verbal	AnnoMI therapist	0.0	100.0	0.0	0.0	100.0	0.0
Sonnet verbal	MFRC	85.4	0.0	0.0	84.5	0.0	0.0
Sonnet verbal	TalkMoves student	0.0	100.0	0.0	0.0	100.0	0.0
Sonnet verbal	TalkMoves teacher	0.0	100.0	0.0	0.0	100.0	0.0

Accepted: mean share of test decisions accepted; none accepted: share of re-splits accepting nothing; error > 5%: share of re-splits whose held-out error exceeds 5%. A re-split that accepts nothing cannot exceed 5%; read exceedance with acceptance.

Table 19: MFRC: corpus LTT over the 16-threshold grid vs one threshold fixed in advance (0.95)

signal	grid of 16: accepted	error	one threshold (0.95): accepted	error
Jev	0.598	0.016	0.170	0.006
OpenAI Decisions	0.000	–	0.000	–
Clef (27B)	0.823	0.030	0.365	0.001
Clef-flash (9B)	0.779	0.027	0.213	0.001
pplx-decider	0.771	0.023	0.656	0.013
Kev-9B	0.671	0.024	0.176	0.010
Kev-27B	0.766	0.021	0.294	0.003
Luna verbal	0.640	0.022	0.745	0.034
Haiku verbal	0.658	0.017	0.549	0.008
gpt-oss verbal	0.000	–	0.718	0.026
Sonnet verbal	0.827	0.025	0.624	0.007

On the four dialogue tasks, with the original whole-cluster split, neither version certifies anything for any coder.

Table 20: LTT on synthetic clustered corpora (300 simulations per row, 30% of clusters hand-coded, $\delta = 0.05$)

clusters in corpus	decisions per cluster	copula loading	within-cluster error correlation	LTT on	P(remainder risk > 5%)	P(population risk > 5%)	mean share accepted
2000	8	0.000	-0.002	decisions	0.103	0.057	0.775
2000	8	0.000	-0.002	i.i.d. clusters	0.000	0.000	0.522
2000	8	0.000	-0.002	corpus	0.000	0.000	0.289
2000	8	0.500	0.070	decisions	0.113	0.067	0.772
2000	8	0.500	0.070	i.i.d. clusters	0.000	0.000	0.477
2000	8	0.500	0.070	corpus	0.000	0.000	0.249
2000	8	0.900	0.364	decisions	0.163	0.137	0.763
2000	8	0.900	0.364	i.i.d. clusters	0.003	0.000	0.362
2000	8	0.900	0.364	corpus	0.003	0.000	0.146
400	10	0.900	0.329	decisions	0.193	0.180	0.616
400	10	0.900	0.329	i.i.d. clusters	0.000	0.000	0.000
400	10	0.900	0.329	corpus	0.000	0.000	0.000
170	30	0.900	0.328	decisions	0.297	0.313	0.656
170	30	0.900	0.328	i.i.d. clusters	0.000	0.000	0.000
170	30	0.900	0.328	corpus	0.000	0.000	0.000

Calibrated confidence conf = 1 – 0.5ûBeta(1, 6); errors correlated within a cluster through a Gaussian copula (the loading is not the error correlation, which is shown). Remainder: the uncoded clusters of the corpus; population: closed form. The best threshold with population risk $\leq 5\%$ accepts 0.841 of decisions.

Table 21: Our corpus LTT procedure: hand-coded units needed before it certifies a coder that is never wrong

A. Grid of 16 thresholds (as used throughout)

task	sampling unit	units in corpus	mean unit weight / B	frozen split certifies it	lower bound	our procedure (median)
MFRC	comment	2000	1.00	yes	119	123
AnnoMI client	transcript	128	0.12	no	unreachable	not reached
AnnoMI client	utterance	1500	0.85	–	119	146.5
AnnoMI therapist	transcript	132	0.13	no	unreachable	not reached
AnnoMI therapist	utterance	1500	0.87	–	119	145
TalkMoves teacher	lesson	444	0.27	no	388	not reached
TalkMoves teacher	sentence	1500	0.78	–	119	163
TalkMoves student	lesson	436	0.08	no	unreachable	not reached
TalkMoves student	sentence	1500	0.90	–	119	139

B. One threshold fixed in advance (0.95)

task	sampling unit	units in corpus	lower bound	our procedure (median)
MFRC	comment	2000	62	64
AnnoMI client	transcript	128	unreachable	not reached
AnnoMI client	utterance	1500	62	75
AnnoMI therapist	transcript	132	unreachable	not reached
AnnoMI therapist	utterance	1500	62	75
TalkMoves teacher	lesson	444	114	not reached
TalkMoves teacher	sentence	1500	62	81.5
TalkMoves student	lesson	436	unreachable	not reached
TalkMoves student	sentence	1500	62	70.5

B: the largest unit's weight. Lower bound: a necessary number over every set of hand-coded units and every visiting order (each unit's bet can multiply the evidence by at most $1 + 0.05 \times$ its weight / B, and certification needs a factor of 320 with the grid, 20 with one threshold); fewer units cannot suffice, and reaching the bound does not guarantee certification. Unreachable: not reached even with every unit but one hand-coded. Our procedure: median number of units at which it first certifies, over 20 random hand-coding sequences (a description of typical runs, not a bound). Transcripts and lessons are the units of our splits; utterances and sentences drawn at random are an alternative design.

Table 22: Dialogue tasks under a different hand-coding design: corpus LTT with single utterances or sentences drawn at random (exploratory; 200 re-splits, 30% hand-coded)

A. Corpus LTT: mean share of the remainder's decisions accepted (percent; re-splits with no certificate count as 0)

signal	AnnoMI client	AnnoMI therapist	TalkMoves teacher	TalkMoves student
Jev	0.0	0.0	0.0	0.0
OpenAI Decisions	0.0	0.0	0.0	0.0
Clef (27B)	0.0	0.0	0.0	0.0
Clef-flash (9B)	0.0	0.0	0.0	0.0
pplx-decider	0.0	0.4	0.0	0.0
Kev-9B	0.0	0.0	0.0	0.0
Kev-27B	0.0	0.2	0.0	0.0
Luna verbal	0.0	0.0	0.0	0.0
Haiku verbal	0.0	0.0	0.0	0.0
gpt-oss verbal	0.0	0.0	0.0	0.0
Sonnet verbal	0.0	0.0	0.0	0.0

B. Corpus LTT: re-splits with any certified threshold

signal	AnnoMI client	AnnoMI therapist	TalkMoves teacher	TalkMoves student
Jev	0 of 200	0 of 200	0 of 200	0 of 200
OpenAI Decisions	0 of 200	0 of 200	0 of 200	0 of 200
Clef (27B)	0 of 200	0 of 200	0 of 200	0 of 200
Clef-flash (9B)	0 of 200	0 of 200	0 of 200	0 of 200
pplx-decider	0 of 200	2 of 200	0 of 200	0 of 200
Kev-9B	0 of 200	0 of 200	0 of 200	0 of 200
Kev-27B	0 of 200	1 of 200	0 of 200	0 of 200
Luna verbal	0 of 200	0 of 200	0 of 200	0 of 200
Haiku verbal	0 of 200	0 of 200	0 of 200	0 of 200
gpt-oss verbal	0 of 200	0 of 200	0 of 200	0 of 200
Sonnet verbal	0 of 200	0 of 200	0 of 200	0 of 200

C. Naive rule: share of re-splits whose accepted decisions exceed 5% error (percent)

signal	AnnoMI client	AnnoMI therapist	TalkMoves teacher	TalkMoves student
Jev	1.0	35.0	15.0	23.0
OpenAI Decisions	6.0	45.5	1.0	67.0
Clef (27B)	70.5	67.5	77.0	67.5
Clef-flash (9B)	74.0	58.5	55.5	38.0
pplx-decider	59.0	51.5	58.0	50.0
Kev-9B	68.0	78.0	48.0	28.0
Kev-27B	72.5	46.5	73.5	57.0
Luna verbal	36.5	19.0	25.5	44.5
Haiku verbal	42.0	31.0	30.0	5.0
gpt-oss verbal	55.5	46.0	22.5	62.0
Sonnet verbal	49.0	37.5	37.5	66.0

Each re-split hand-codes a uniformly random 30% of the task's segments, with their context as the coder saw it; the uncoded segments are the remainder, and the sampling unit is the segment, so B is the largest segment weight. Dependence among segments of one transcript is allowed. Coders tested: the seven decision models and Luna's, Haiku's, gpt-oss's and Sonnet's verbalized confidence (44 coder–task pairs). Corpus LTT's accepted remainder decisions exceeded 5% error in 0% of re-splits in every cell. Panel C: the naive rule on the same re-splits, for comparison.

C.5 EQ3: ties at the top and per-code recalibration

To find what limits triage (§5.3), we measured how many decisions each coder places at its single highest confidence and how many of those are wrong, then fitted a separate map per predicted code. The per-code maps improve ranking for every coder with enough data; the category-only control shows how much of that gain comes from knowing which code was predicted, without any confidence at all.

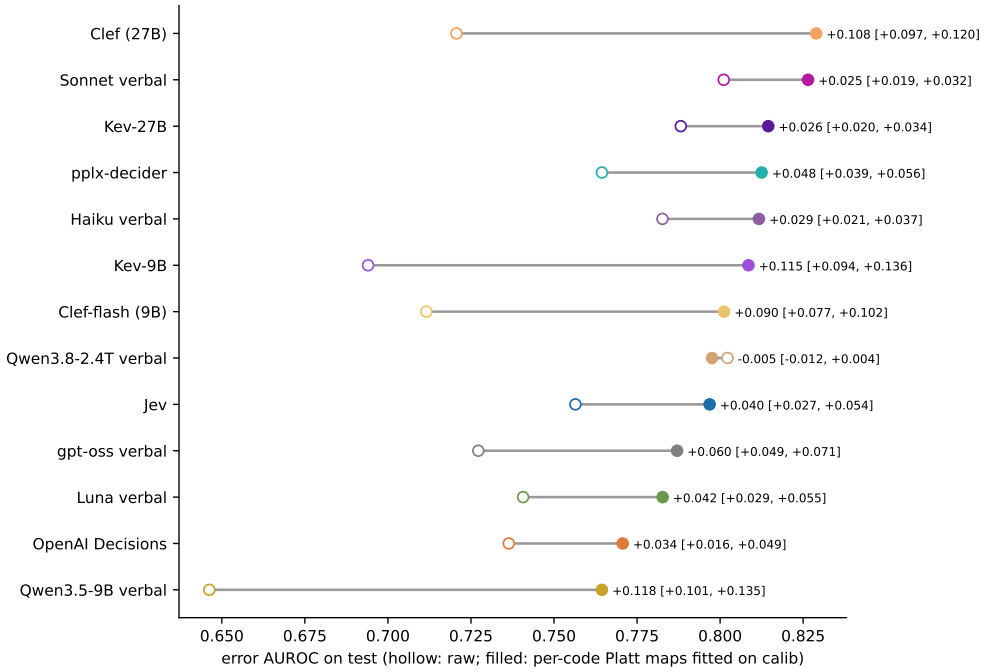


Figure 11: EQ3: error AUROC with raw confidence (hollow) and with per-code Platt maps fitted on the calibration split (filled), pooled over tasks, with cluster-bootstrap 95% CIs for the gain, conditional on the fitted maps. Qwen3.8-2.4T ran on a subset whose calibration split is too small for per-code maps.

Table 23: Top-confidence tie blocks on the dialogue tasks (test split)

task	signal	top confidence	decisions at top	share at top	error at top
AnnoMI client	Jev	1.000	274	0.302	0.123
AnnoMI client	OpenAI Decisions	1.000	176	0.198	0.121
AnnoMI client	Clef (27B)	0.979	1	0.001	0.000
AnnoMI client	pplx-decider	0.998	1	0.001	0.000
AnnoMI client	Kev-27B	0.995	1	0.001	0.000
AnnoMI client	Luna verbal	1.000	6	0.007	0.000
AnnoMI client	Haiku verbal	0.990	1	0.001	0.000
AnnoMI client	Sonnet verbal	0.970	16	0.017	0.000
AnnoMI therapist	Jev	1.000	214	0.225	0.068
AnnoMI therapist	OpenAI Decisions	1.000	117	0.131	0.027
AnnoMI therapist	Clef (27B)	0.963	2	0.002	0.000
AnnoMI therapist	pplx-decider	0.998	1	0.001	0.000
AnnoMI therapist	Kev-27B	0.996	1	0.001	0.000
AnnoMI therapist	Luna verbal	1.000	72	0.074	0.073
AnnoMI therapist	Haiku verbal	0.990	4	0.004	0.000
AnnoMI therapist	Sonnet verbal	0.980	15	0.013	0.000
TalkMoves teacher	Jev	1.000	104	0.082	0.083
TalkMoves teacher	OpenAI Decisions	1.000	271	0.249	0.107
TalkMoves teacher	Clef (27B)	0.948	1	0.000	0.000
TalkMoves teacher	pplx-decider	0.997	1	0.001	1.000
TalkMoves teacher	Kev-27B	0.978	1	0.001	0.000
TalkMoves teacher	Luna verbal	1.000	9	0.009	0.039
TalkMoves teacher	Haiku verbal	0.980	3	0.002	0.000
TalkMoves teacher	Sonnet verbal	0.960	36	0.037	0.000
TalkMoves student	Jev	1.000	119	0.106	0.065
TalkMoves student	OpenAI Decisions	1.000	9	0.009	0.000
TalkMoves student	Clef (27B)	0.945	1	0.000	0.000
TalkMoves student	pplx-decider	0.995	1	0.001	0.000
TalkMoves student	Kev-27B	0.983	1	0.001	0.000
TalkMoves student	Luna verbal	1.000	11	0.011	0.088
TalkMoves student	Haiku verbal	0.990	5	0.005	0.000
TalkMoves student	Sonnet verbal	0.960	16	0.015	0.000

Table 24: A3 (exploratory): per-code recalibration of confidence (error AUROC on test, pooled)

comparison	AUROC a	AUROC b	a – b	95% CI low	95% CI high
Jev: per-code calibrated - raw	0.797	0.756	0.040	0.027	0.054
OpenAI Decisions: per-code calibrated - raw	0.771	0.736	0.034	0.016	0.049
Clef (27B): per-code calibrated - raw	0.829	0.721	0.108	0.097	0.120
Clef-flash (9B): per-code calibrated - raw	0.801	0.712	0.090	0.077	0.102
pplx-decider: per-code calibrated - raw	0.813	0.764	0.048	0.039	0.056
Kev-9B: per-code calibrated - raw	0.809	0.694	0.115	0.094	0.136
Kev-27B: per-code calibrated - raw	0.814	0.788	0.026	0.020	0.034
Luna verbal: per-code calibrated - raw	0.783	0.741	0.042	0.029	0.055
Haiku verbal: per-code calibrated - raw	0.812	0.783	0.029	0.021	0.037
gpt-oss verbal: per-code calibrated - raw	0.787	0.727	0.060	0.049	0.071
Qwen3.5-9B verbal: per-code calibrated - raw	0.764	0.646	0.118	0.101	0.135
Sonnet verbal: per-code calibrated - raw	0.826	0.801	0.025	0.019	0.032
Qwen3.8-2.4T verbal: per-code calibrated - raw	0.798	0.802	-0.005	-0.012	0.004
Jev per-code (full calib) - Sonnet verbal raw	0.797	0.801	-0.004	-0.018	0.008
Jev raw - Sonnet verbal raw	0.756	0.801	-0.045	-0.060	-0.032
Jev per-code - Sonnet verbal per-code (own calib)	0.797	0.826	-0.030	-0.041	-0.019
Jev per-code (full calib) - Qwen3.8-2.4T verbal raw	0.790	0.802	-0.013	-0.048	0.019
Jev raw - Qwen3.8-2.4T verbal raw	0.769	0.802	-0.033	-0.069	-0.001
Jev per-code - Qwen3.8-2.4T verbal per-code (own calib)	0.790	0.798	-0.008	-0.043	0.024

Per-code Platt maps fitted on calib (codes with ≥ 50 calib decisions; others use the task map). CIs resample test clusters with the fitted maps held fixed, so they are conditional on those maps.

Table 25: Error AUROC from raw confidence, from the predicted code alone, and from per-code maps (pooled)

signal	raw confidence	predicted code only	per-code maps
Clef (27B)	0.721	0.707	0.829
Clef-flash (9B)	0.712	0.693	0.801
Haiku verbal	0.783	0.696	0.812
Jev	0.756	0.714	0.797
Kev-27B	0.788	0.657	0.814
Kev-9B	0.694	0.713	0.809
Luna verbal	0.741	0.694	0.783
OpenAI Decisions	0.736	0.659	0.771
gpt-oss verbal	0.727	0.692	0.787
pplx-decider	0.764	0.691	0.813
Sonnet verbal	0.801	0.696	0.826

'Predicted code only' scores each decision by the calibration accuracy of its predicted code (multi-label: of its code), with no confidence.

C.6 EQ5: configurations and cost

We compared single coders, two-stage cascades, agreement gates and a seven-model committee, each at the API cost of its own policy (§5.5). The first table pools the naive and oracle rules over the five tasks; the second gives what corpus LTT certifies on MFRC, with the share of comments a person still reads; the third tests selected differences between configurations.

Table 26: A3 (exploratory): configurations under the naive and oracle rules, pooled over the five tasks

configuration	naive: accepted	naive: error	naive: USD/1k	oracle: accepted	oracle: USD/1k
Jev → human	32.9	6.8	0.021	16.3	0.008
OpenAI Decisions → human	22.7	4.7	0.046	17.8	0.065
Clef (27B) → human	24.0	5.8	0.213	20.3	0.213
Clef-flash (9B) → human	25.4	7.0	0.080	20.7	0.080
pplx-decider → human	40.8	7.6	0.013	32.1	0.013
Kev-9B → human	23.9	6.3	0.073	19.2	0.058
Kev-27B → human	44.2	7.0	0.315	37.2	0.315
Decision-model committee (7) → human	37.5	7.2	0.849	30.9	0.849
pplx-decider + Kev-27B agree → human	42.7	6.8	0.334	40.5	0.334
Jev + Kev-27B agree → human	42.9	7.0	0.362	36.2	0.362
Kev-27B → Sonnet verbal → human	49.6	6.9	1.406	42.8	1.328
Luna verbal → human	20.2	4.6	0.082	16.2	0.066
Haiku verbal → human	16.1	3.2	1.137	16.8	1.906
Sonnet verbal → human	43.4	6.2	1.700	40.1	1.964
Jev → Haiku verbal → human	33.4	6.7	0.579	17.9	1.741
Haiku verbal → Jev → human	31.6	6.4	1.155	16.9	1.913
Jev + Haiku verbal agree → human	26.0	5.5	1.158	28.4	1.954
Jev + Luna verbal agree → human	34.3	6.6	0.130	17.8	0.131
Jev → Sonnet verbal → human	40.4	6.2	1.327	39.2	1.746
Jev + Sonnet verbal agree → human	43.1	6.5	1.736	39.1	2.011
Haiku verbal + Sonnet verbal agree → human	41.5	6.0	3.870	38.2	3.870

Accepted: percent of decisions auto-accepted; error: percent wrong among accepted decisions (tasks weighted equally); USD/1k: API cost per 1,000 segments under that policy. 'A + B agree' accepts where both give the same label (confidence = the lower).

Table 27: A3 (exploratory): configurations certified by corpus LTT on MFRC (percent; the only task where anything is certified when whole clusters are hand-coded)

configuration	certified: accepted	error	USD per 1k comments	remainder comments still read	all comments still read	overall error vs adjudicated
Jev → human	59.8	1.6	0.042	93.2	95.2	1.1
OpenAI Decisions → human	0.0	–	0.000	100.0	100.0	0.2
Clef (27B) → human	82.3	3.1	0.245	73.3	81.4	2.6
Clef-flash (9B) → human	77.9	2.7	0.092	76.2	83.4	2.2
pplx-decider → human	77.1	2.3	0.029	73.2	81.3	1.9
Kev-9B → human	67.1	2.4	0.073	99.9	99.9	1.8
Kev-27B → human	76.6	2.1	0.315	84.6	89.3	1.7
Decision-model committee (7) → human	84.5	2.9	0.970	68.2	77.8	2.5
pplx-decider + Kev-27B agree → human	81.1	2.6	0.344	74.9	82.5	2.2
Jev + Kev-27B agree → human	75.7	2.2	0.357	78.1	84.7	1.8
Kev-27B → Sonnet verbal → human	82.3	2.5	4.141	74.0	81.9	2.1
Luna verbal → human	64.0	2.2	0.218	93.0	95.1	1.5
Haiku verbal → human	65.8	1.7	3.473	83.5	88.5	1.2
Sonnet verbal → human	82.7	2.5	4.521	65.7	76.1	2.2
Jev → Haiku verbal → human	66.5	1.7	3.513	83.1	88.2	1.3
Haiku verbal → Jev → human	66.5	1.8	3.509	83.1	88.2	1.3
Jev + Haiku verbal agree → human	63.3	1.8	3.516	90.2	93.2	1.3
Jev + Luna verbal agree → human	67.9	2.2	0.260	88.4	91.9	1.6
Jev → Sonnet verbal → human	77.0	1.9	4.479	78.4	85.0	1.5
Jev + Sonnet verbal agree → human	73.5	2.1	4.563	82.5	87.8	1.6
Haiku verbal + Sonnet verbal agree → human	74.2	2.3	7.995	75.3	82.8	1.8

Multi-stage configurations are certified jointly over the pipeline. Accepted, error and remainder comments still read refer to the uncoded remainder (the test split); a comment is still read if at least one of its decisions is left for a person. All comments still read adds the 608 hand-coded calibration comments to the numerator and denominator. Overall error vs adjudicated: certified decisions scored against the adjudicated reference, the rest taken as human-coded; human-only coding of MFRC has 0.2% error against that reference.

Table 28: A3 (exploratory): paired configuration contrasts, accepted share a – b in percentage points [95% CI]

A. Naive rule and oracle, pooled over the five tasks

a	b	naive	oracle
Jev → Luna verbal	Luna verbal	+13.0 [-0.7, +18.4]	+0.4 [-0.9, +10.8]
Jev → Haiku verbal	Haiku verbal	+17.3 [+0.9, +19.1]	+1.1 [-0.9, +9.6]
Jev → Haiku verbal	Jev	+0.5 [+0.0, +10.2]	+1.6 [+0.6, +17.2]
Jev → Haiku verbal	OpenAI Decisions → Haiku verbal	+9.2 [-5.3, +15.5]	-1.9 [-9.1, +5.7]
Jev	OpenAI Decisions	+10.3 [-5.0, +18.7]	-1.5 [-9.2, +4.9]
Jev + Haiku verbal agree	Jev → Haiku verbal	-7.4 [-11.3, +9.6]	+10.5 [-6.1, +10.3]
Jev + Haiku verbal agree	Jev	-6.9 [-10.0, +12.7]	+12.1 [-0.2, +14.7]
Jev + Haiku verbal agree	Haiku verbal	+10.0 [-1.6, +18.6]	+11.7 [-4.3, +12.0]
Jev + Luna verbal agree	Jev	+1.3 [-3.4, +13.4]	+1.5 [-2.0, +11.7]
Jev → Sonnet verbal	Sonnet verbal	-3.0 [-4.5, +2.3]	-0.9 [-2.7, +0.6]
Jev	Sonnet verbal	-10.5 [-28.3, -5.4]	-23.8 [-28.0, -12.1]
Jev + Haiku verbal agree	Sonnet verbal	-17.3 [-25.3, -5.7]	-11.7 [-19.7, -6.1]
Jev + Sonnet verbal agree	Sonnet verbal	-0.2 [-8.9, +3.1]	-1.0 [-4.6, +3.3]
Kev-27B	Jev	+11.3 [+6.3, +27.2]	+20.9 [+11.1, +27.6]
Decision-model committee (7)	Jev	+4.6 [+1.6, +21.1]	+14.6 [+5.2, +20.2]
Decision-model committee (7)	Sonnet verbal	-5.9 [-13.9, +1.6]	-9.2 [-12.8, -0.5]
pplx-decider + Kev-27B agree	Sonnet verbal	-0.7 [-9.5, +7.1]	+0.4 [-6.9, +8.7]

B. Corpus LTT on MFRC

a	b	corpus LTT	USD per 1k, a – b
Jev → Luna verbal	Luna verbal	-4.8 [-64.6, +0.2]	0.04
Jev → Haiku verbal	Haiku verbal	+0.7 [-4.1, +8.1]	0.04
Jev → Haiku verbal	Jev	+6.7 [-3.1, +25.2]	3.47
Jev → Haiku verbal	OpenAI Decisions → Haiku verbal	+4.3 [-4.0, +62.7]	-0.13
Jev	OpenAI Decisions	+59.8 [+42.9, +70.1]	0.04
Jev + Haiku verbal agree	Jev → Haiku verbal	-3.2 [-6.9, +1.7]	0.00
Jev + Haiku verbal agree	Jev	+3.5 [-6.6, +20.1]	3.47
Jev + Haiku verbal agree	Haiku verbal	-2.5 [-3.4, +3.4]	0.04
Jev + Luna verbal agree	Jev	+8.1 [-1.8, +23.3]	0.22
Jev → Sonnet verbal	Sonnet verbal	-5.7 [-9.3, +0.6]	-0.04
Jev	Sonnet verbal	-22.9 [-42.3, -13.1]	-4.48
Jev + Haiku verbal agree	Sonnet verbal	-19.4 [-23.5, -12.5]	-1.01
Jev + Sonnet verbal agree	Sonnet verbal	-9.2 [-16.1, -7.5]	0.04
Kev-27B	Jev	+16.7 [+7.1, +38.2]	0.27
Decision-model committee (7)	Jev	+24.6 [+14.2, +40.9]	0.93
Decision-model committee (7)	Sonnet verbal	+1.8 [-8.3, +2.8]	-3.55
pplx-decider + Kev-27B agree	Sonnet verbal	-1.6 [-10.0, +3.8]	-4.18

1,000 clustered bootstrap replicates with thresholds refitted; related tasks share cluster draws. Naive and oracle: equal-weight means over the five tasks. Corpus LTT: MFRC only, the one task where it certifies anything when whole clusters are hand-coded. Every configuration ends with a person coding what it does not accept; 'A + B agree' accepts where both give the same label.

C.7 EQ6: validity checks

Four checks bound specific threats to the results (§5.6). Repeatability sends identical requests twice; prompt sensitivity tries a second elicitation prompt for the LLM baselines; adjudication re-examines sampled disagreements between coders and the human labels; and the memorization probe, with its decoy-name control, tests whether the LLMs reproduce the public datasets. None of them proves validity; each addresses a particular alternative explanation. The model snapshot records exactly what was called.

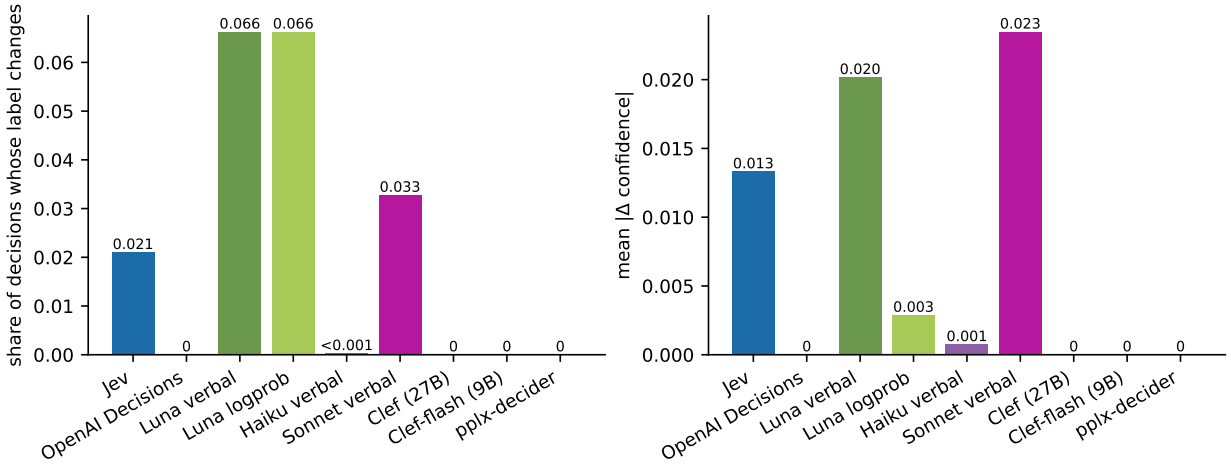


Figure 12: EQ6: run-to-run repeatability. Identical requests were sent twice for 100 test segments per task. Left: share of decisions whose label changes. Right: mean absolute change in confidence. 0: no observed change in these repeats; <0.001: a nonzero value below display precision.

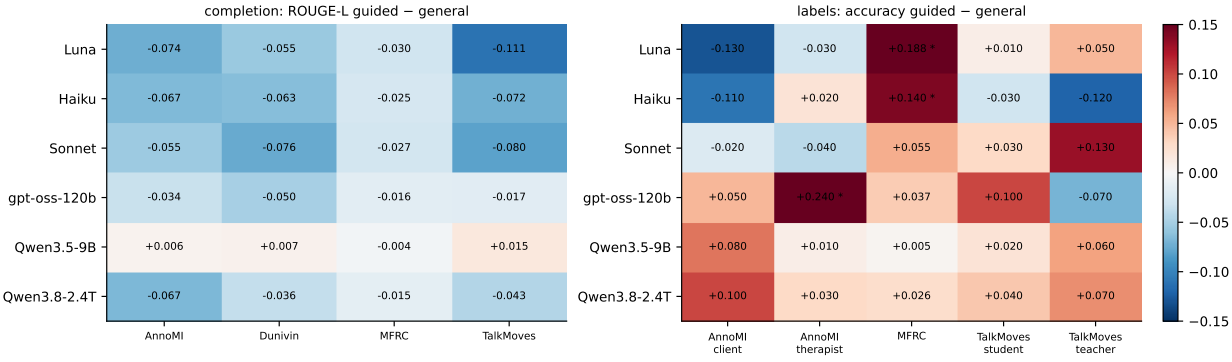


Figure 13: EQ6: memorization probe. Change in completion overlap (left, ROUGE-L) and label recall (right, accuracy) when the dataset is named, relative to an unnamed prompt. Asterisks mark cells flagged by the prespecified rule; Table 35 gives the decoy control.

Table 29: Run-to-run repeatability (identical requests sent twice; 100 test segments per task; mean over tasks)

A. How much the outputs change

signal	segments	decisions	label flip rate	share identical	mean $\lvert\Delta p\rvert$	mean $\lvert\Delta$ confidence$\rvert$
Jev	100	240	0.021	0.612	0.008	0.013
OpenAI Decisions	100	240	0	1.000	0.000	0.000
Luna verbal	100	240	0.066	0.567	0.030	0.020
Luna logprob	100	240	0.066	0.455	0.031	0.003
Haiku verbal	100	240	<0.001	0.989	0.001	0.001
Sonnet verbal	100	240	0.033	0.514	0.016	0.023
Clef (27B)	100	240	0	1.000	0.000	0.000
Clef-flash (9B)	100	240	0	1.000	0.000	0.000
pplx-decider	100	240	0	1.000	0.000	0.000

B. Accuracy in each run, and where the changes fall

signal	accuracy, run 1	accuracy, run 2	flips at low confidence (AUROC)
Jev	0.743	0.741	0.967
OpenAI Decisions	0.723	0.723	–
Luna verbal	0.742	0.742	0.874
Luna logprob	0.742	0.742	0.647
Haiku verbal	0.702	0.703	0.895
Sonnet verbal	0.764	0.765	0.921
Clef (27B)	0.693	0.693	–
Clef-flash (9B)	0.685	0.685	–
pplx-decider	0.759	0.759	–

Flips at low confidence: AUROC of low confidence for the decisions whose label changed (– : no changes).

Table 30: LLM prompt sensitivity: verbalized distribution vs canonical prompt (amendment v1.4)

A. Error ranking

task	signal	decisions	err. AUROC canonical	err. AUROC distribution	difference	lo	hi
MFRC	Sonnet verbal	1200	0.933	0.897	-0.036	-0.052	-0.019
MFRC	Haiku verbal	1200	0.942	0.897	-0.045	-0.063	-0.029
AnnoMI client	Sonnet verbal	150	0.750	0.726	-0.024	-0.074	0.018
AnnoMI client	Haiku verbal	150	0.733	0.776	0.043	-0.041	0.136
AnnoMI therapist	Sonnet verbal	150	0.808	0.837	0.029	-0.012	0.083
AnnoMI therapist	Haiku verbal	150	0.770	0.717	-0.053	-0.139	0.014
TalkMoves teacher	Sonnet verbal	150	0.817	0.747	-0.070	-0.135	0.000
TalkMoves teacher	Haiku verbal	150	0.747	0.767	0.021	-0.080	0.120
TalkMoves student	Sonnet verbal	150	0.829	0.758	-0.071	-0.133	-0.012
TalkMoves student	Haiku verbal	150	0.786	0.770	-0.015	-0.082	0.046
Pooled	Sonnet verbal	1800	0.827	0.793	-0.034	-0.056	-0.011
Pooled	Haiku verbal	1800	0.796	0.786	-0.010	-0.045	0.022

B. Agreement and calibration

task	signal	decisions	κ canonical	κ distribution	ECE canonical	ECE distribution
MFRC	Sonnet verbal	1200	0.346	0.365	0.048	0.031
MFRC	Haiku verbal	1200	0.166	0.174	0.128	0.076
AnnoMI client	Sonnet verbal	150	0.413	0.432	0.065	0.079
AnnoMI client	Haiku verbal	150	0.479	0.453	0.118	0.158
AnnoMI therapist	Sonnet verbal	150	0.751	0.715	0.053	0.083
AnnoMI therapist	Haiku verbal	150	0.634	0.661	0.155	0.158
TalkMoves teacher	Sonnet verbal	150	0.590	0.603	0.089	0.045
TalkMoves teacher	Haiku verbal	150	0.554	0.528	0.141	0.215
TalkMoves student	Sonnet verbal	150	0.539	0.548	0.118	0.090
TalkMoves student	Haiku verbal	150	0.445	0.430	0.261	0.330
Pooled	Sonnet verbal	1800	0.528	0.533	0.074	0.066
Pooled	Haiku verbal	1800	0.456	0.449	0.161	0.187

Same test items (first 150 per task). Difference = distribution – canonical error AUROC, paired cluster bootstrap 95% CI; related tasks share cluster draws.

Table 31: Who was right when a coder disagreed with the human label (adjudicated sample)

coder	verdicts	n	coder right, human wrong	human right, coder wrong	both defensible (codebook under-specified)	neither right
Jev	all	117	0.598	0.325	0.068	0.009
Jev	LLM panel	90	0.622	0.289	0.089	0.000
Jev	author	27	0.519	0.444	0.000	0.037
Luna verbal	all	118	0.644	0.288	0.059	0.008
Luna verbal	LLM panel	84	0.667	0.250	0.083	0.000
Luna verbal	author	34	0.588	0.382	0.000	0.029
Haiku verbal	all	118	0.585	0.314	0.085	0.017
Haiku verbal	LLM panel	89	0.573	0.292	0.112	0.022
Haiku verbal	author	29	0.621	0.379	0.000	0.000
gpt-oss verbal	all	141	0.553	0.390	0.043	0.014
gpt-oss verbal	LLM panel	108	0.556	0.380	0.056	0.009
gpt-oss verbal	author	33	0.545	0.424	0.000	0.030

Shares of adjudicated disagreements on a high-confidence-enriched sample drawn from six label sources (Jev in both formats, Luna, Haiku, gpt-oss and Sonnet's subset run); the four coders shown have enough disagreements to break down. The panel resolved the items on which its two members agreed; the author resolved 36 items the panel disputed plus a 12-item audit of panel-agreed items (deviations 2–3).

Table 32: Memorization probe: completing the second half of dataset items, overlap with the true continuation (amendment v1.3)

model	dataset	items	ROUGE-L, named	ROUGE-L, unnamed	difference	p (Bonferroni)
Haiku	annomi	100	0.079	0.146	-0.067	1.000
Haiku	dunivin	100	0.080	0.144	-0.063	1.000
Haiku	mfrc	100	0.080	0.105	-0.025	1.000
Haiku	talkmoves	100	0.082	0.155	-0.072	1.000
Luna	annomi	100	0.082	0.156	-0.074	1.000
Luna	dunivin	100	0.095	0.151	-0.055	1.000
Luna	mfrc	100	0.082	0.111	-0.030	1.000
Luna	talkmoves	100	0.063	0.174	-0.111	1.000
gpt-oss-120b	annomi	100	0.057	0.090	-0.034	1.000
gpt-oss-120b	dunivin	100	0.016	0.066	-0.050	1.000
gpt-oss-120b	mfrc	100	0.038	0.055	-0.016	1.000
gpt-oss-120b	talkmoves	100	0.049	0.066	-0.017	1.000
Qwen3.5-9B	annomi	100	0.142	0.137	0.006	1.000
Qwen3.5-9B	dunivin	100	0.143	0.137	0.007	1.000
Qwen3.5-9B	mfrc	100	0.098	0.102	-0.004	1.000
Qwen3.5-9B	talkmoves	100	0.163	0.148	0.015	1.000
Qwen3.8-2.4T	annomi	100	0.083	0.150	-0.067	1.000
Qwen3.8-2.4T	dunivin	100	0.112	0.148	-0.036	1.000
Qwen3.8-2.4T	mfrc	100	0.086	0.101	-0.015	1.000
Qwen3.8-2.4T	talkmoves	100	0.125	0.168	-0.043	1.000
Sonnet	annomi	100	0.098	0.153	-0.055	1.000
Sonnet	dunivin	100	0.099	0.175	-0.076	1.000
Sonnet	mfrc	100	0.089	0.116	-0.027	1.000
Sonnet	talkmoves	100	0.076	0.156	-0.080	1.000

Named: the dataset is named and the exact original text requested. Unnamed: plain continuation. Difference: named – unnamed mean ROUGE-L; p: one-sided Wilcoxon signed-rank test for named > unnamed, Bonferroni-adjusted. Near-verbatim matches are in Table 33.

Table 33: Memorization probe: near-verbatim reproduction (counts of items, amendment v1.3)

model	dataset	items	near-verbatim items, named	near-verbatim items, unnamed	longest run, named (words)	flagged
Haiku	annomi	100	0	0	3	no
Haiku	dunivin	100	0	0	2	no
Haiku	mfrc	100	0	0	3	no
Haiku	talkmoves	100	0	1	3	no
Luna	annomi	100	0	0	2	no
Luna	dunivin	100	1	0	45	no
Luna	mfrc	100	0	0	2	no
Luna	talkmoves	100	0	0	4	no
gpt-oss-120b	annomi	100	0	0	2	no
gpt-oss-120b	dunivin	100	0	0	1	no
gpt-oss-120b	mfrc	100	0	0	2	no
gpt-oss-120b	talkmoves	100	0	0	3	no
Qwen3.5-9B	annomi	100	0	0	6	no
Qwen3.5-9B	dunivin	100	0	0	8	no
Qwen3.5-9B	mfrc	100	0	0	4	no
Qwen3.5-9B	talkmoves	100	1	1	10	no
Qwen3.8-2.4T	annomi	100	0	0	4	no
Qwen3.8-2.4T	dunivin	100	0	1	8	no
Qwen3.8-2.4T	mfrc	100	0	0	4	no
Qwen3.8-2.4T	talkmoves	100	0	0	5	no
Sonnet	annomi	100	0	0	3	no
Sonnet	dunivin	100	0	1	9	no
Sonnet	mfrc	100	0	0	3	no
Sonnet	talkmoves	100	0	0	2	no

Near-verbatim: a run of ≥ 10 identical words covering $\geq 50\%$ of the true continuation; counts of the 100 items. Longest run: the longest identical run under the named prompt only. Flagged: the pre-specified rule (Bonferroni $p < 0.05$ for named > unnamed in Table 32, or ≥ 2 near-verbatim items under the named prompt). Matches include a predictable counting sequence (TalkMoves) and Du Bois quotations (Dunivin).

Table 34: Memorization probe: recalling annotated labels without the codebook (amendment v1.3)

model	task	items	accuracy guided	accuracy general	difference	p (Bonferroni)	evidence of label leakage
Haiku	AnnoMI client	100	0.510	0.620	-0.110	1.000	no
Haiku	AnnoMI therapist	100	0.580	0.560	0.020	1.000	no
Haiku	MFRC	100	0.314	0.174	0.140	0.014	yes
Haiku	TalkMoves student	100	0.440	0.470	-0.030	1.000	no
Haiku	TalkMoves teacher	100	0.160	0.280	-0.120	1.000	no
Luna	AnnoMI client	100	0.580	0.710	-0.130	1.000	no
Luna	AnnoMI therapist	100	0.530	0.560	-0.030	1.000	no
Luna	MFRC	100	0.408	0.220	0.188	0.015	yes
Luna	TalkMoves student	100	0.530	0.520	0.010	1.000	no
Luna	TalkMoves teacher	100	0.520	0.470	0.050	1.000	no
gpt-oss-120b	AnnoMI client	100	0.560	0.510	0.050	1.000	no
gpt-oss-120b	AnnoMI therapist	100	0.680	0.440	0.240	0.004	yes
gpt-oss-120b	MFRC	100	0.260	0.223	0.037	1.000	no
gpt-oss-120b	TalkMoves student	100	0.510	0.410	0.100	0.621	no
gpt-oss-120b	TalkMoves teacher	100	0.200	0.270	-0.070	1.000	no
Qwen3.5-9B	AnnoMI client	100	0.720	0.640	0.080	1.000	no
Qwen3.5-9B	AnnoMI therapist	100	0.570	0.560	0.010	1.000	no
Qwen3.5-9B	MFRC	100	0.174	0.170	0.005	1.000	no
Qwen3.5-9B	TalkMoves student	100	0.420	0.400	0.020	1.000	no
Qwen3.5-9B	TalkMoves teacher	100	0.360	0.300	0.060	1.000	no
Qwen3.8-2.4T	AnnoMI client	100	0.720	0.620	0.100	0.959	no
Qwen3.8-2.4T	AnnoMI therapist	100	0.730	0.700	0.030	1.000	no
Qwen3.8-2.4T	MFRC	100	0.185	0.160	0.026	1.000	no
Qwen3.8-2.4T	TalkMoves student	100	0.490	0.450	0.040	1.000	no
Qwen3.8-2.4T	TalkMoves teacher	100	0.280	0.210	0.070	1.000	no
Sonnet	AnnoMI client	100	0.720	0.740	-0.020	1.000	no
Sonnet	AnnoMI therapist	100	0.730	0.770	-0.040	1.000	no
Sonnet	MFRC	100	0.573	0.518	0.055	1.000	no
Sonnet	TalkMoves student	100	0.590	0.560	0.030	1.000	no
Sonnet	TalkMoves teacher	100	0.630	0.500	0.130	0.287	no

Accuracy against the majority label (Jaccard for MFRC). Guided names the dataset and asks what its annotators assigned.

Table 35: Flagged label-probe cells: decoy-name control and coding accuracy on 'recalled' items

A. Label recall with the real name, a decoy name and no name

model	task	items	real name	decoy name	no name	real – decoy	p (real > decoy)	p (Bonferroni, 3 cells)
Haiku	MFRC	100	0.314	0.246	0.174	0.068	0.021	0.063
Luna	MFRC	100	0.408	0.280	0.220	0.128	0.012	0.035
gpt-oss-120b	AnnoMI therapist	100	0.680	0.650	0.440	0.030	0.304	0.911

B. Coding accuracy on 'recalled' and other items

model	task	recalled items	other items	coding accuracy, recalled	coding accuracy, other	p (recalled > other)
Haiku	MFRC	22	78	0.841	0.782	0.085
Luna	MFRC	35	65	0.868	0.883	0.779
gpt-oss-120b	AnnoMI therapist	33	67	0.970	0.672	0.000

Decoy: the same dataset description with a fictitious name (deviation 14). Memory is supported only if the real name beats the decoy. Recalled items: guided better than no name. Coding accuracy is the model's RQ1 accuracy on those items.

Table 36: Model snapshot: what was called, which version answered, and when

coder	provider	model requested	version returned	calls (all arms)	first call (UTC)	last call (UTC)
Jev	typesafe	jev-1.13.0	jev-1.13.0	65995	2026-10-10	2026-10-11
OpenAI Decisions	openai decisions	gpt-6-luna	gpt-6-luna	65993	2026-10-11	2026-10-11
Clef	cloudflare	@cf/cloudflare/clef	clef	14617	2026-10-11	2026-10-11
Clef-flash	cloudflare	@cf/cloudflare/clef-flash	clef-flash	65993	2026-10-11	2026-10-11
pplx-decider	perplexity	pplx-decider-v1.1-27b	pplx-decider-v1.1-27b	65993	2026-10-11	2026-10-11
Kev-9B	kev	jaredpalmer/kev-9b	kev-latest	51368	2026-10-11	2026-10-11
Kev-27B	kev	jaredpalmer/kev-27b	kev-latest	14117	2026-10-11	2026-10-11
Luna	openai	gpt-6-luna	gpt-6-luna	13905	2026-10-10	2026-10-11
Haiku	anthropic	claude-haiku-4-5	claude-haiku-4-5-20251001	11280	2026-10-10	2026-10-12
Sonnet	anthropic	claude-sonnet-5-5	claude-sonnet-5-5	12437	2026-10-10	2026-10-12
gpt-oss-120b	together	openai/gpt-oss-120b	openai/gpt-oss-120b	10030	2026-10-10	2026-10-11
Qwen3.5-9B	together	Qwen/Qwen3.5-9B	Qwen/Qwen3.5-9B	9930	2026-10-11	2026-10-11
Qwen3.8-2.4T	together	Qwen/Qwen3.8-2.4T-A95B	Qwen/Qwen3.8-2.4T-A95B	3281	2026-10-11	2026-10-11

Several APIs were in public beta (OpenAI Decisions from 2026-10-06; decision models launched September–October 2026). Kev weights: Hugging Face revisions db029f0 (kev-9b) and af0e6d5 (kev-27b), the latest on the call dates; kev package commit fc4a17e. Results describe these versions on these dates; the released harness allows re-running against later versions.

D Requests

LLM prompt (canonical; AnnoMI client format; the exchange shown is invented for illustration, since AnnoMI's text is not redistributed). The system prompt carries the codebook; option letters and their order are shared with the decision-model request.

```
SYSTEM:
You are an experienced qualitative researcher applying a fixed codebook (deductive coding). Apply the
  codebook exactly as written.

Task: In this motivational interviewing session, what type of talk is the client's utterance with respect to
  the target behaviour change given in ‘setting‘?
Unit of analysis: one client utterance.

Options (exactly one applies):
A) Change talk
   Definition: The client's own language in favor of the change goal.
   Applies when:
   - Commitment: an agreement, intention, or obligation regarding future change.
   - Taking steps: concrete steps the client has recently taken to make a change.
   - Reason: rationale, basis, justification, or motive to make a change.
   - Other: other statements that clearly reflect an intention to change.
B) Sustain talk
   Definition: The client's language giving reasons against changing or for maintaining the status quo.
   Applies when:
   - Commitment: an agreement, intention, or obligation not to change.
   - Taking steps: concrete steps the client has recently taken away from change.
   - Reason: rationale, basis, justification, or motive not to change.
   - Other: other statements that clearly reflect an intention not to change.
C) Neutral talk
   Definition: No indication of client inclination toward or away from change.
   Applies when:
   - Follow/neutral: responses that show no inclination toward or away from change.
   - Ask: asking for clarification or information.

Respond with JSON only. "rationale": one sentence explaining your decision. "answer": the letter of the
  option that applies. "probability": an integer from 0 to 100, your probability that this answer is
  correct.

USER:
setting: Motivational interviewing session. Target behaviour change: reducing alcohol consumption.

previous_turn (context only, not coded):
Therapist: What do you notice about how the drinking and your mood go together?

segment:
Honestly, when I'm low I reach for a drink, and afterwards I usually feel worse.

Judge only the client utterance in ‘segment‘; ‘previous_turn‘ is context and is not coded.
```

Decision-model request (the same item). Jev, Clef, pplx-decider and Kev receive this exact JSON. OpenAI Decisions receives the same content mapped to its choice and predicate types.

```
{
 "state": {
  "setting": "Motivational interviewing session. Target behaviour change: reducing alcohol consumption.",
  "previous_turn": "Therapist: What do you notice about how the drinking and your mood go together?",
  "segment": "Honestly, when I'm low I reach for a drink, and afterwards I usually feel worse."
 },
 "questions": {
  "code": {
   "type": "choice",
   "instructions": "In this motivational interviewing session, what type of talk is the client's utterance
  with respect to the target behaviour change given in ‘setting‘? Judge only the client utterance in
  ‘segment‘; ‘previous_turn‘ is context and is not coded.",
   "criteria": {
    "Change talk": {
     "definition": "The client's own language in favor of the change goal.",
```

```
    "applies_when": [
     "Commitment: an agreement, intention, or obligation regarding future change.",
     "Taking steps: concrete steps the client has recently taken to make a change.",
     "Reason: rationale, basis, justification, or motive to make a change.",
     "Other: other statements that clearly reflect an intention to change."
    ]
   },
   "Sustain talk": {
    "definition": "The client's language giving reasons against changing or for maintaining the status
  quo.",
    "applies_when": [
     "Commitment: an agreement, intention, or obligation not to change.",
     "Taking steps: concrete steps the client has recently taken away from change.",
     "Reason: rationale, basis, justification, or motive not to change.",
     "Other: other statements that clearly reflect an intention not to change."
    ]
   },
   "Neutral talk": {
    "definition": "No indication of client inclination toward or away from change.",
    "applies_when": [
     "Follow/neutral: responses that show no inclination toward or away from change.",
     "Ask: asking for clarification or information."
    ]
   }
  }
 }
}
```

E Memorization probe

For each of MFRC, AnnoMI, TalkMoves and Dunivin we drew 100 items of at least 30 words (20 for TalkMoves), with a fixed seed. The model saw the first half of the words (plus the previous turn for dialogue). The *guided* prompt named the dataset and its citation and asked for the exact original continuation. The *general* prompt asked only for a continuation [19]. We scored word-level ROUGE-L [35] and the longest common run of words, and called a completion near-verbatim if the run was at least 10 words and covered at least half of the true continuation. The label probe gave the item and the bare code labels, without definitions. The decoy control replaced the dataset name and citation with a fictitious one while keeping the description. Tables 32–35 report all cells.